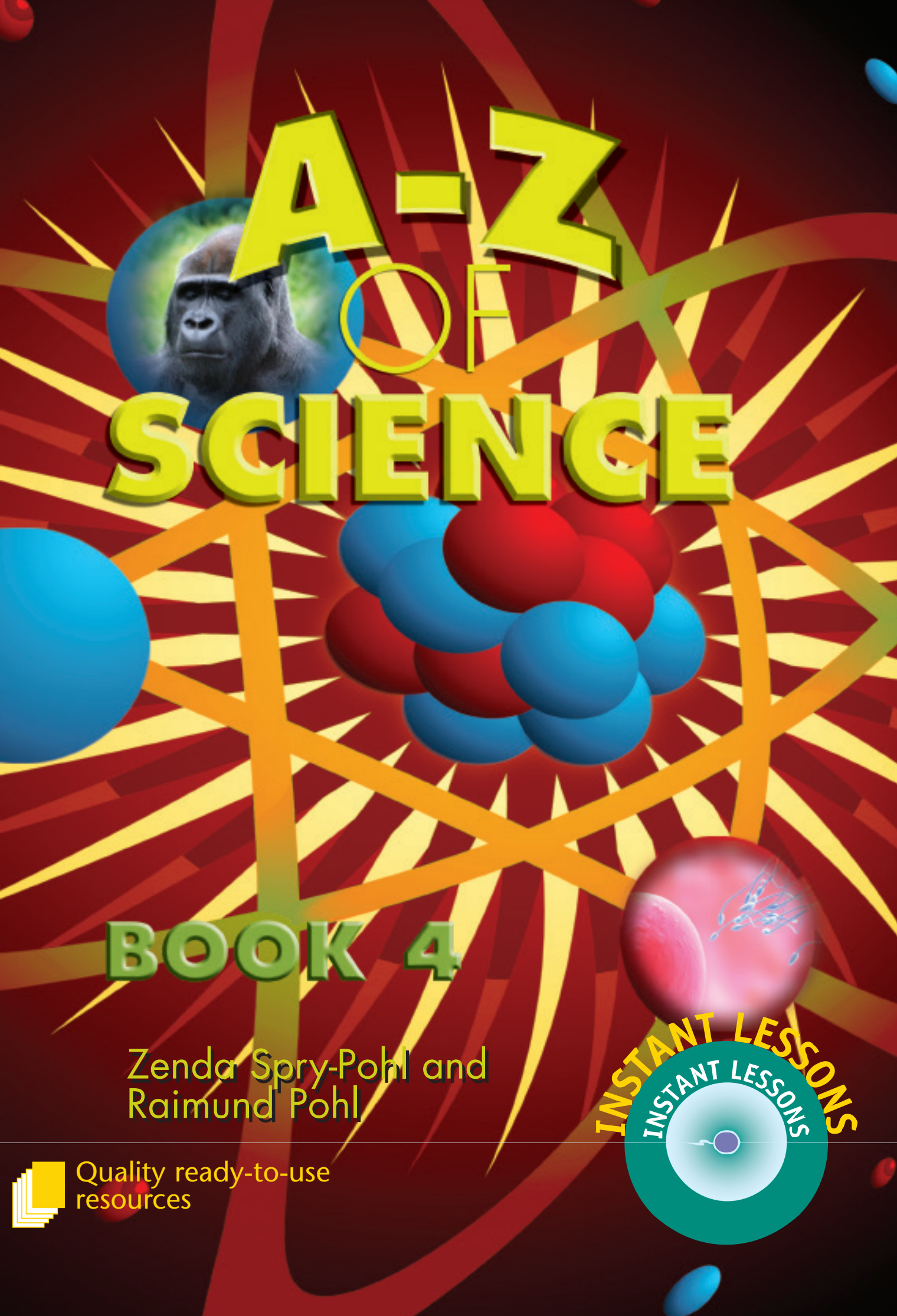


A-Z OF SCIENCE

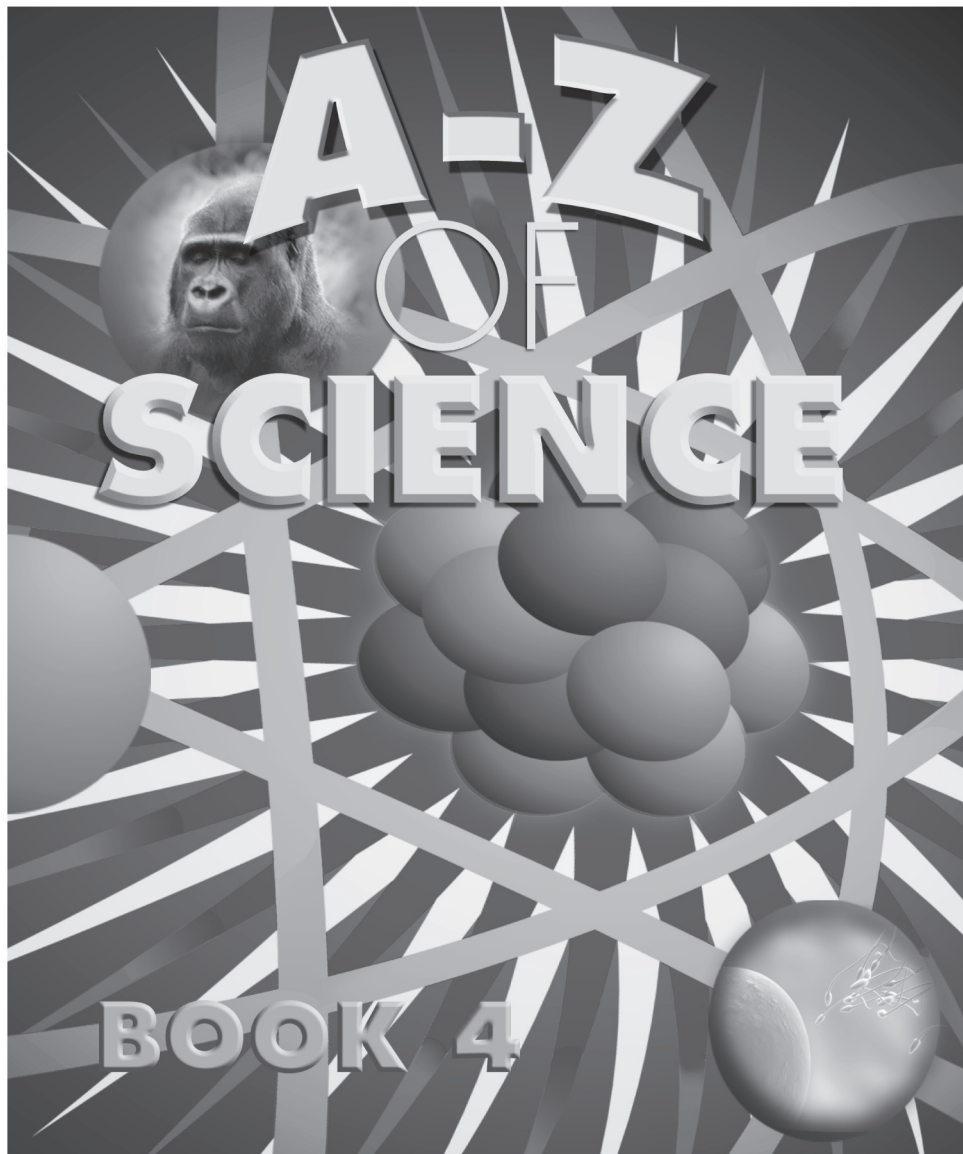


BOOK 4

Zenda Spry-Pohl and
Raimund Pohl

 Quality ready-to-use
resources





Zenda Spry-Pohl and
Raimund Pohl

This book was commissioned by Jennifer Gough, production was by Piper Print Media, Gail McManus Graphics designed the cover and Boris Silvestri supplied the illustrations.

© Zenda Spry-Pohl, Raimund Pohl and Blake Education 2008

Copying of this book for educational purposes

The purchasing educational institution may only photocopy pages within this book in accordance with The Australian Copyright Act 1968 (the Act) and provided the educational institution (or body that administers it) has given a remuneration notice to the Copyright Agency Limited (CAL) under the Act. It is a breach of copyright to scan or in any other way make digital copies of the worksheets.

It is mandatory that ALL photocopies are recorded by the institution for CAL survey purposes.

Copyright Agency Limited
Level 15, 233 Castlereagh Street
Sydney NSW 2000
Telephone: (02) 9394 7600
Facsimile: (02) 9394 7601
E-mail: enquiry@copyright.com.au

The Act does not permit the purchasing educational institution to store any part of this book in a retrieval system, or transmit any part of it in any form by any means without the prior written approval of the publisher. All enquiries should be made to the publisher at the address below.

Printed in Australia by Ligare Book Printer

Blake
EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au

ISBN 978 1 92571 004 5

Introduction

This series of photocopiable books extends across all branches of this subject area, including biology, chemistry, physics, astronomy, earth and environmental science. The activities include basic revision of fundamental concepts as well as ideas and tasks that will challenge students to think beyond recalling facts. Each topic is designed to engage student interest in science, as well as consolidate and extend their scientific knowledge and understanding. Book 3 is suitable for years 7 and 8, while Book 4 is more advanced and will suit years 9 and 10. Teachers will find these worksheets useful in a variety of ways, including homework, assessment, revision and extension. They are designed to suit both independent and group work, and are also convenient ways of targeting the learning of individual students. For some worksheets, students may find textbooks or research materials useful.

Contents

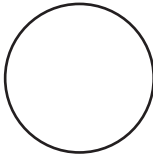
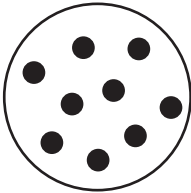
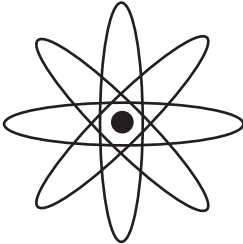
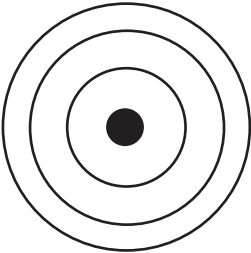
Atoms – models, atomic structure, compound formation, ions, ionic bonding, electronic configuration.....	4
Biotechnology – uses of, timeline, benefits and disadvantages	6
Chemical reactions – types of reactions, word equations, balancing equations	7
DNA – history, discovery, structure, amino acid codes, replication, mutation	9
Earthquakes – causes, wave types, triangulation, seismogram interpretation	11
Electricity – use, making electricity, earthing, conductors, circuits, resistance.....	13
Fossils – fossil types, fossilisation process, law of superposition, correlation	15
Hormones and nerves – what hormones are and do, endocrine glands, central nervous system	16
Indicators and pH – definitions and uses.....	18
Joules, watts, amps and volts – discovery, relationship of joules to power, watts, amps and volts	19
Motion – speed, acceleration, velocity, relationship to Newton's Law, graphing.....	20
Natural events – volcanoes, cyclones, global effects.....	22
Newton's Laws – Newton's three laws.....	23
Non-renewable resources – types, formation, use	25
Nuclear fission and fusion – definitions, equations, radioactivity, isotopes.....	26
Periodic Table – structure, identification of elements by group and row, electronic configuration, bonding.....	28
Quasars and stars – what they are, star formation, redshift, star lifecycle.....	29
Reflection and refraction – definitions, how they occur, types of mirrors and lenses.....	31
Tsunamis – what they are and how they occur, relationship of earthquake magnitude to amplitude.....	33
Universe formation – evidence for formation, three theories, Doppler effect.....	35
Volcanoes – types, how they occur, eruption related to volcanic type, formation of the atmosphere, graphing.....	36
Waves – properties of waves, energy carriers, uses, absorption	38
X-rays and other rays – discovery, uses, electromagnetic spectrum, gamma rays, alpha and beta particles, equations.....	40
Y chromosome – chromosome discovery, Punnet squares, diseases, graphing, mutations.....	42
Zygotes and reproduction – types of reproduction, human reproductive systems, zygote to foetus.....	44
Selected answers	46

Atoms

There are four distinct models for representing atoms: the billiard ball model, the plum pudding model, the planetary model and the shell model.

John Dalton in 1803 developed the concept that pure substances were made of indivisible solid particles called atoms. John Joseph Thomson in 1897 discovered negatively charged particles that he called electrons. His model was the plum pudding model. In 1911, Ernest Rutherford discovered the small positive nucleus, which was surrounded by electrons. Niels Bohr in 1913 proposed that the electrons were arranged in layers around the nucleus and that this arrangement accounted for the chemical and physical properties of the atom.

1 Name each atomic model shown and identify the scientist who developed it. Then label the electron orbits and the nucleus in the atomic models.

NAME OF MODEL				
DIAGRAM OF MODEL				
SCIENTIST				

All of the chemicals you use are formed when atoms react with each other. Knowing how atoms are organised is the basis of chemistry and enables you to make your own compounds.

The atomic number of an atom is found by counting the number of protons in the nucleus. Protons are electrically positive particles. The number of protons equals the number of electrons in an electrically balanced atom, so an element with an atomic number of 5 has 5 protons and 5 electrons. The neutral particles in the nucleus are called neutrons. The number of neutrons and protons makes up the mass number of the element. Electrons are arranged in shells. The first shell can have only two electrons. The second, third and fourth shells can have up to 8 electrons.

2 Mark each statement as true or false. Then correct the false statements so that they are true.

- a** The number of neutrons makes up the atomic number of an element. _____
- b** The mass number is made up of protons and neutrons. _____
- c** Electrons are found in shells surrounding the nucleus. _____

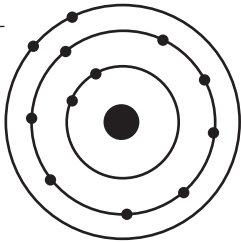
3 Using the shell model, draw these atoms.

2 protons 2 neutrons 2 electrons	6 protons 6 neutrons 6 electrons	8 protons 8 neutrons 8 electrons	15 protons 15 neutrons 15 electrons

How ionic compounds form

Electrons are electrically negative. Atoms with 1 electron or 7 electrons in the outer shell are very reactive. Atoms can lose or gain electrons, which results in an atom having either a positive electrical charge or a negative electrical charge. These charged atoms are called ions. This transfer of electrons usually occurs between a metal and a non-metal. For example, the electronic configuration of sodium (Na) is 2.8.1 and for chlorine (Cl) it is 2.8.7. When these atoms react with each other, the sodium loses its outer shell electron to chlorine. Sodium loses 1 electron and chlorine gains 1 electron from sodium. This makes sodium electrically positive and chlorine electrically negative, and now both atoms have 8 electrons in their outermost shells. As the electrical charges are opposite, they form a compound called sodium chloride. The compound is held together by electrostatic attraction. These types of compounds are called ionic compounds.

- 4 Mark each statement as true or false. Then correct the false statements so that they are true.
- a Only atoms that have lost electrons are called ions.
 - b All elements in group 1 form ionic compounds when combined with group 7 elements.
 - c All electrons are positively charged.
 - d Ionic compounds are held together by electrostatic charge.
 - e Each shell surrounding an atom contains up to 8 electrons, except the first shell.



- 5 The magnesium atom is shown at the right. Copy it into your workbook, and then draw the ion of magnesium 2.8, the atom of fluorine 2.8.7 and fluorine’s ion.

Every element’s atomic number in its neutral state has the same number of protons and electrons. The electrons are placed into shells. The number of shells an atom has relates to the period (or row) of the Periodic Table that it is in.

The name and atomic number of an atom of the first twenty elements are as shown in the box below.

• hydrogen 1	• boron 5	• fluorine 9	• aluminium 13	• chlorine 17
• helium 2	• carbon 6	• neon 10	• silicon 14	• argon 18
• lithium 3	• nitrogen 7	• sodium 11	• phosphorus 15	• potassium 19
• beryllium 4	• oxygen 8	• magnesium 12	• sulfur 16	• calcium 20

- 6 Using the information above, figure out the name, atomic number and period of atoms with the electronic configurations shown in the table.

ELECTRONIC CONFIGURATION	NAME	ATOMIC NUMBER	PERIOD
2.4			
2.8.1			
2.8.6			
2.8.8.1			

- 7 Given the atomic mass numbers below, figure out the number of protons, electrons and neutrons for each one. Then write the name of the element.

MASS NUMBER	NO. PROTONS	NO. NEUTRONS	NO. ELECTRONS	NAME OF ELEMENT
23	11			
27			13	
14		7		
40	18			
9			4	

Biotechnology

Biotechnology is the use of living organisms or other biological systems in a technological field. Uses of biotechnology include the manufacture of drugs, environmental management such as waste recycling, the use of genetically engineered bacteria to produce human hormones and the production of genetically modified crops.

- 1 Show the historical development of biotechnology by writing the underlined words in each bullet point beside the correct date in the table at the right. Use your logic – some developments had to occur before the next one could be invented; for example, the microscope had to be invented before we could see bacteria.

+ Brewing beer, fermenting wine, <u>baking bread</u> with the help of yeast	+ In the early 90s, the FDA in the US approved the <u>first GM food</u> called the ‘Flavr Savr’ tomato.
+ Recently, GloFish, the <u>first biotech pet</u> , hit the North American market.	+ Biotechnology was characterised by <u>recombinant DNA</u> technology in the 80s.
+ More recently, <u>monoclonal antibody work</u> is developing a treatment for multiple sclerosis.	+ In the 70s, the <u>DNA composition of chimpanzees and gorillas</u> was discovered to be 98% similar to that of humans.
+ British scientists, led by Ian Wilmut from the Roslin Institute, report <u>cloning a sheep named Dolly</u> using DNA from two adult sheep cells.	+ James Watson and Francis Crick described the structure of <u>deoxyribonucleic acid</u> , called DNA for short.
+ After cloning was successful, a ‘rough draft’ of the human genome was mapped in the <u>Human Genome Project</u> .	+ Between World War 1 and 2, Alexander Fleming noticed that a certain mould could stop the development of bacteria, leading to the <u>first antibiotic</u> , penicillin.
+ The <u>microscope</u> was invented by Zacharias Janssen.	

DATE	DEVELOPMENT
Before 8000 BC	Collecting of seeds for replanting.
Around 7000 BC	
1590	
1862	Louis Pasteur discovers the bacterial origin of fermentation.
1928	
1953	
1972	
1980	
1994	
1997	
2000	
2003	
2007	

A case study

Cotton is a very popular crop and it suffers badly from insect attack. Insect-resistant genetically engineered Bt cotton has been developed to reduce pesticide use. However, in the long term farmers will still be faced with pests becoming resistant to pesticides. It appears that yields from GM crops such as canola are generally higher, so a farmer may earn more money from the same crop. Gene technology has the potential to develop plants that are more nutritious and yield bigger harvests than conventional plants.

- 2 Organise the information above about GM plants into a table showing benefits and problems. Include other facts you already know and can research.

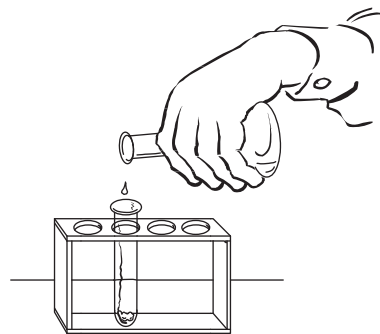
Genetically modified plants

BENEFITS	PROBLEMS

Chemical reactions

All life depends on chemical reactions – for example, the photosynthesis in plants and the digestion of the food you eat. There are several different types of reactions. Every reaction requires a balanced number of reactants and products. Remember that matter cannot be created or destroyed, which is why we need the correct and balanced number of atoms as reactants and products. In symbol reactions, the following symbols are used to show the state of the reactant or product:

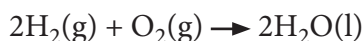
- » (s) = solid
- » (l) = liquid
- » (g) = gas; in a water solution it is written (aq) = aqueous



Synthesis reactions occur when two different reactants form a new product, for example:

hydrogen gas + oxygen gas $\rightarrow$ water

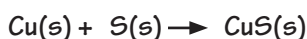
Written chemically, the reaction looks like this:



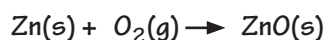
Note that the number 2 in front of H_2O and H_2 in the example above is a multiplier of all the atoms in the compound or molecule. Including this number balances the equation.

1 Complete the following word reactions and balance the equations.

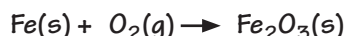
a copper + sulphur $\rightarrow$ copper _____



b zinc + oxygen $\rightarrow$ _____



c iron + _____ $\rightarrow$ iron III oxide



Decomposition reactions are reactions where a compound breaks down into simpler compounds or elements.

2 Complete the following word reactions and balance the equations.

a water (liquid) + electricity $\rightarrow$ hydrogen gas + _____



b zinc sulphide + oxygen $\rightarrow$ zinc _____ + sulfur dioxide



c lithium carbonate $\rightarrow$ lithium oxide + carbon _____



Displacement reactions are also called single replacement reactions. In these reactions, one element or a group of elements replaces another element or group of elements in a compound.

3 Complete the following word reactions and balance the equations.

a zinc + copper _____ $\rightarrow$ copper + zinc sulfate



b chlorine + potassium bromide → bromine + potassium chloride



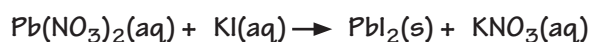
c sodium + water → sodium hydroxide + _____



Precipitate reactions or double replacement reactions are those reactions where two compounds form an insoluble salt called a precipitate. Many of these reactions provide pigments for products such as paints.

4 Complete the following word reactions and balance the equations.

a lead nitrate + potassium iodide → lead iodide + _____



b silver nitrate + sodium chloride → sodium nitrate + silver chloride



c sodium hydroxide + copper sulfate → sodium sulfate + copper hydroxide



Acid reactions with a metal or a metal compound make a salt and other products. For example:

hydrochloric acid + sodium → sodium chloride + hydrogen gas

Note that 'chloric' becomes chloride. Similarly with sulfuric acid, the name of the compound becomes sulfate. Nitric acid produces a salt called nitrate. Ethanoic acid produces a salt called ethanoate. Metal carbonates produce carbon dioxide when reacted with an acid.

5 Complete the following word equations.

a nitric acid + magnesium → magnesium nitrate + _____

b sulfuric acid + copper carbonate → copper sulfate + water + _____

c hydrochloric acid + lithium oxide → lithium chloride + _____

Acids react with bases (hydroxides), usually to form a salt and water. These reactions are called **neutralisation reactions** because the acids and bases form water and are neither acidic nor basic.

6 Complete the following word equations.

a sulfuric acid + potassium hydroxide → _____ + water

b hydrochloric acid + sodium hydroxide → _____ + water

7 Name the type of reaction occurring in each equation.

a $3\text{Fe}(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{Fe}_3\text{O}_4(\text{s}) + 4\text{H}_2(\text{g})$ _____

b $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ _____

c $2\text{Na}(\text{s}) + \text{F}_2(\text{g}) \rightarrow 2\text{NaF}(\text{s})$ _____

d $\text{PbO}(\text{s}) + \text{H}_2(\text{g}) \rightarrow \text{Pb}(\text{s}) + \text{H}_2\text{O}(\text{g})$ _____

e $\text{Mg}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$ _____

f $\text{KOH}(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{KCl}(\text{aq}) + \text{H}_2\text{O}(\text{s})$ _____

DNA

DNA has a pattern for every living organism and is specific to an individual, for example, you. DNA can be used to determine relationships such as the parentage of a child, and to provide clues at crime scenes. DNA molecules are made of smaller chemical units called nucleotides. Without the nucleotides, there would be no DNA nor would you exist.

DNA has all the instructions to make and shape your body. DNA stands for deoxyribonucleic acid, and it was discovered in the 1860s by Friedrich Miescher. However, it wasn't considered important until Rosalind Franklin demonstrated through X-ray crystallography photographs that DNA had a helical structure. Watson and Crick pursued the concept and looked at the structural function of DNA. They showed that DNA was also a method of copying genetic information.

- 1 Using an example, explain the advantage of exact DNA replication.

DNA is found in the nucleus of the cell, which is the cell's control centre. Chromatin has genetic material and is composed of DNA and a protein. At certain stages of the cell's life, the chromatin condenses into chromosomes prior to replication of the cell.

- 2 In the box at the right, draw a diagram of a cell showing the nucleus and chromosomes that it contains.

Inside the double helix structure of DNA are the bases adenine (A), thymine (T), cytosine (C) and guanine (G). Adenine and thymine form two hydrogen bonds to each other while guanine and cytosine form three hydrogen bonds to each other. This means that A always bonds with T, and C with G.

The outer structure is made from sugar-phosphate units. The sugars are connected to the four bases.

- 3 Complete the following sequence to show the corresponding strand of DNA.

AAT	CGT	CAT	TAG	TGA	GAT
TTA	G__	__	__	__	__

DNA carries a template for producing a large variety of proteins from only 20 amino acids. The DNA controls the specific sequence of amino acids to produce the required proteins.

As the pieces of DNA code are assembled in a particular sequence, they control how specific amino acids are joined together. Different amino acid combinations make different proteins.

- 4 Using the DNA code from the tables on the next page, identify the amino acid represented by each of these codes.

a AGA_____

b TAG_____

c TTC_____

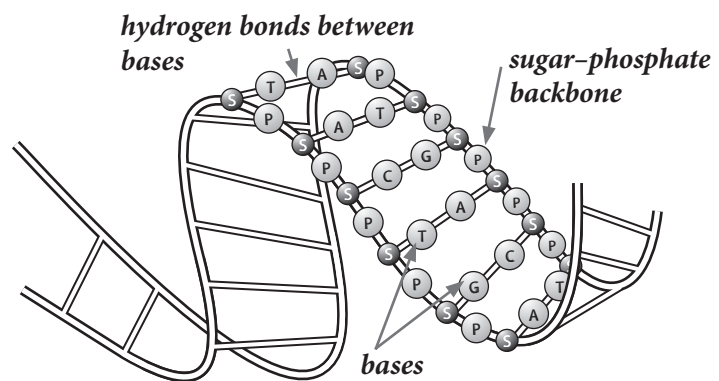
d ATA_____

e TTG_____

f GTC_____

g CAA_____

h TGG_____



DNA CODE	AMINO ACID	DNA CODE	AMINO ACID	DNA CODE	AMINO ACID
AAA AAG	Phenylalanine	CAA CAG	Valine	GTT GTC	Glutamine
AAT AAC	Leucine	TGA TGG	Threonine	TTA TTG	Asparagine
AGA AGG	Serine	CGA CGG	Alanine	TTT TTC	Lysine
TAA TAG	Isoleucine	ATA ATG	Tyrosine	CCA CCC	Glycine

When DNA replicates, it unzips itself and forms two single chromatids. The chromatids then replicate and join up to make a copy of the original DNA chromatin strand. When body cells divide (mitosis), the DNA must replicate exactly. If it fails to do so, the cell either dies or produces a cell that can be beneficial or has characteristics that make it less successful in its environment.

5 Draw four simplified diagrams to show how DNA unzips and replicates itself. The first one has been done for you.

Diagram 1

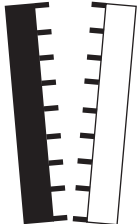


Diagram 2

Diagram 3

Diagram 4

In sexual reproduction (meiosis), the chromosome number is halved to form the ovum and sperm cells. Variation in the chromosomes can occur when parts of chromosomes sometimes swap during meiosis.

Mutations

Mutations are permanent changes in the DNA. Some mutations are beneficial, while others may produce detrimental effects. These mutations to the DNA can be caused by external factors such as X-rays, radiation from the Sun or close proximity to radioactive materials and chemicals like PCBs.

6 Using examples, explain the advantages and disadvantages of DNA mutating.

7 Describe two specific situations in which radiation or chemicals cause DNA mutations.

8 Explain how DNA controls cellular activity.

Earthquakes

The edges of tectonic plate boundaries are rough and jagged and have many faults in them. As the plates move against each other, they get stuck and pressure on the boundary or a block increases. When the force of a block or plate boundary is suddenly released the stored energy radiates out in all directions. The energy is released as seismic waves. These waves shake the ground and anything on it!

Earthquakes generate compression, transverse and longitudinal waves – that is, P, S and L waves. The amplitude or height of the S waves determines the magnitude of an earthquake. Scientists calculate the magnitude using a time–velocity graph. The magnitude is the measure of the amount of energy released by the earthquake.

The time is calculated by subtracting the P wave from the S wave as they arrive at the seismic station, and is measured in seconds. The amplitude of the S wave is measured in millimetres.

1 Define these terms in the context of earthquakes using your own words

a Magnitude _____

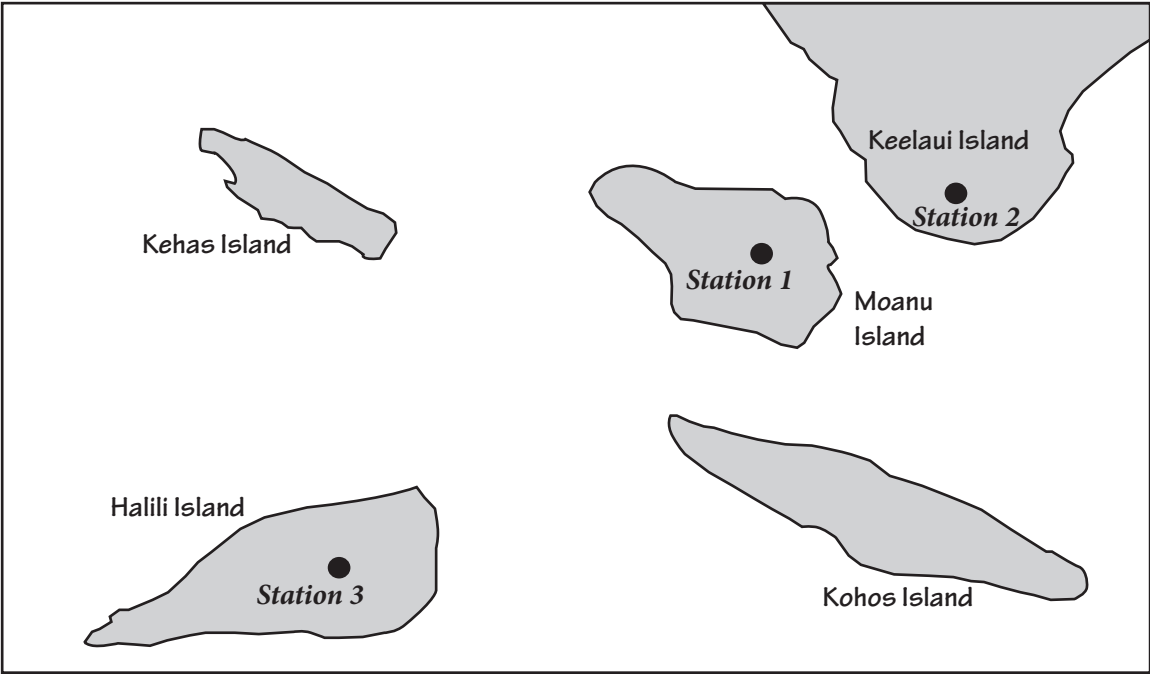
b Seismic wave _____

c Amplitude _____

The strength of the earthquake is based on a logarithmic scale – the Richter scale. An increase of one unit in the Richter scale represents a ten-fold increase in the amplitude of the seismic wave. Using the information from different seismic stations, the location and depth of an earthquake can be calculated. This process is called triangulation.

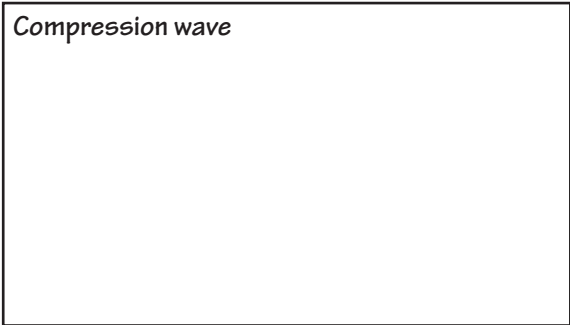
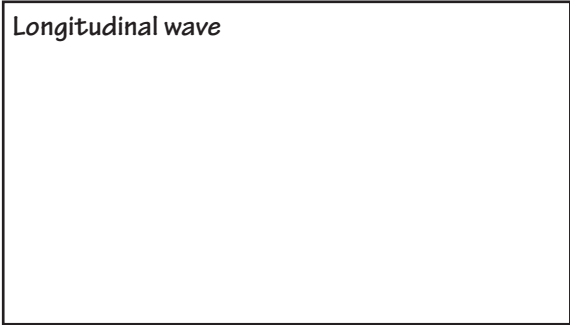
2 Triangulation exercise. Using the scale 1 cm = 1000 km, calculate the radius of the circle from each station. Then draw the three circles on the map below to find the epicentre.

SEISMIC STATION	TIME DIFFERENCE BETWEEN S AND P WAVES IN SECONDS	DISTANCE FROM EPICENTRE (km)	RADIUS OF CIRCLE
1	160	5 000	
2	280	7 000	
3	322	13 800	



P waves are compression waves and will travel through liquids, but S waves or transverse waves do not. Longitudinal waves (L waves) move from side to side and R or Rayleigh waves are rolling waves.

3 Draw a longitudinal wave and a compression wave.

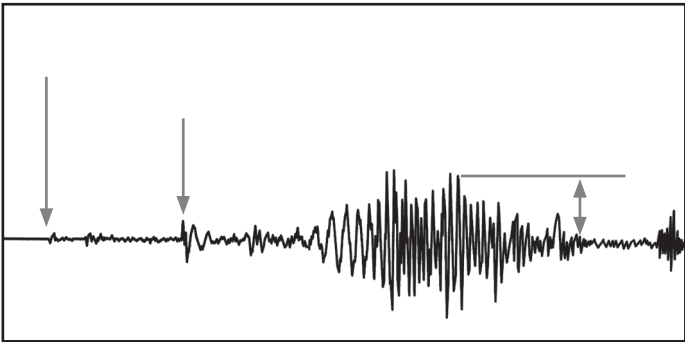


4 Label the seismogram at the right with these terms: P wave, S wave, Amplitude.

P waves and S waves help scientists to determine the structure of the Earth and the rocks that are within the Earth’s crust. Each rock type has specific properties, which affect the generation, reflection and refraction of the seismic waves.

L waves and R waves can sometimes be seen on the surface if the magnitude of the earthquake is high.

The epicentre of an earthquake is the location on the surface of an earthquake. The focus or hypocentre is the location in the ground where the earthquake occurs.



5 Answer the following by placing a ruler on the given values on the diagram at the right. The intersecting line will give you the result you need.

a Derive the magnitude and the distance of the epicentre of the earthquake that has an amplitude of 75 mm and a difference between P and S waves of 10 seconds.

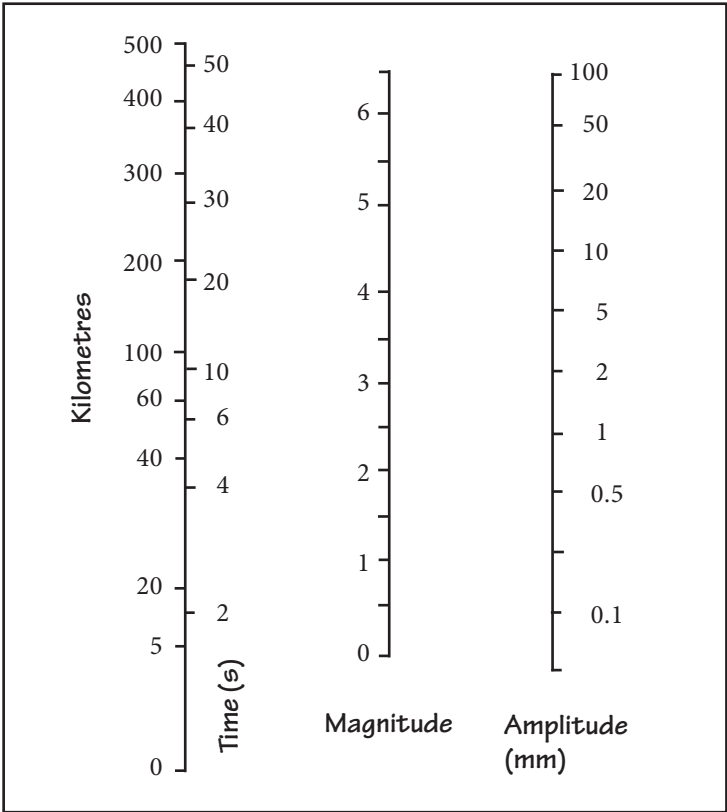
b Derive the time taken for a distance of 400 km and an amplitude of 0.2 mm.

c Derive the magnitude if the time is 2 seconds and the amplitude is 25 mm.

d Derive the distance, given the magnitude of 6.4 and the time of 10 seconds.

6 Present all the information on this worksheet so that a 10-year-old student will understand it. Use the scaffold provided below to structure the information.

- + Introduction
- + How earthquakes are generated
- + The differences in the types of waves
- + How earthquakes are measured
- + Conclusion



Electricity

Charge

Electricity is a form of energy that has physical properties related to the force and motion of electrically charged particles – the electrons. Neutral objects have equal numbers of protons and electrons.

1 Use these terms to label the diagrams below: -ve charge, +ve charge, neutral, polarised.

+ + + + - - - - - -	- - - - + + + + + + + + + +	+ + + + + + + + - - - -	+ + + + + + + + - - - - - -
---	---	---	---

Electricity is made up of two kinds of electric charge: +ve (positive) and -ve (negative). Gaining or losing electrons can charge objects; this can happen through friction, contact or induction.

2 Use the terms in the box to complete the table.

+ friction	+ electric toothbrush	+ static from car seat
+ induction charge	+ contact	+ light switch

	+ + + + + + + + + + + + + +	+ + + + + + + + + + + +	+ + + + + + + + cloth
TYPE			
EXAMPLE			

Earth

In order to make household wiring safe, excess electrons have to flow into the ground – that is, be earthed. An electrical ‘earth’ is anything that can give up or take on an unlimited amount of charge, such as the ground. When an object becomes charged, -ve (negative) electrons are added or removed.

3 Describe two events which could occur at your house that would have to be earthed.

A Van de Graaff generator produces static electricity that makes your hair stand on end. It has a rubber belt, a motor to turn the belt, a metal plate and a metal dome.

4 Briefly describe how this generator can make your hair stand on end.

Conductors

In metals, electrons can ‘flow’ from one atom to another atom. A flow of electrical charge makes an electric current. Electricity flows only through conductors such as metals. Plastics and glass are electrical insulators.

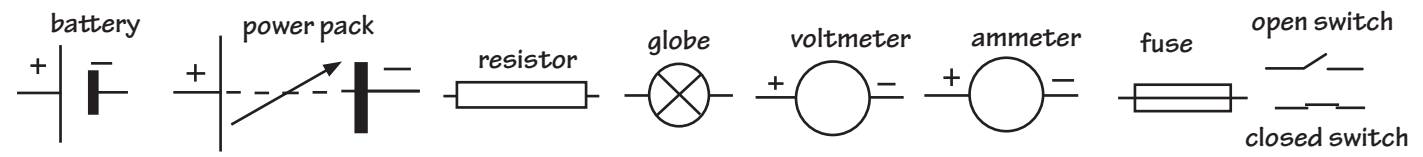
5 List five specific examples of conductors and five non-conductors.

a conductors

b non-conductors

Circuits

An electrical energy source such as a battery provides a force that can move an electric charge through a circuit. Electrical diagrams use the following symbols. The flow of electrons always goes from negative to positive, and wires are always shown as straight lines.

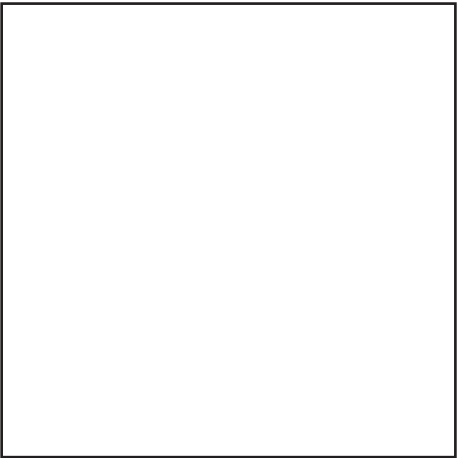


- 6

a

In the box at the right, draw and label a circuit diagram in which a globe will light up. Include these items: battery, wires, globe, a fuse, switch.
- b

Briefly explain how your circuit works and label it to show the direction of the flow of electrons.



When globes are connected **in series** the current must flow through the complete circuit. If a globe is disconnected, the circuit is ‘broken’ and no current can flow.

When globes or resistors are connected **in parallel**, they function independently of each other. Circuits in houses, cars, offices and most party lights are wired in parallel.

- 7

Using the circuit you drew above as a guide, in your workbook draw a circuit with three globes connected in series, and a second circuit with three globes connected in parallel.
- 8

Describe the advantages and disadvantages of series circuits and parallel circuits.

	ADVANTAGES	DISADVANTAGES
Series circuits		
Parallel circuits		

Voltmeters measure the ‘driving force energy’ or ‘potential difference’ in volts. In a circuit, they are connected in parallel. Ammeters measure the ‘current’ flowing through a circuit in amperes or amps. They are connected in series.

Resistance is measured in units called ohms (Ω). Ohm’s Law states: *When a steady current is flowing through a resistor, the current flowing is proportional to the potential difference.* This means that the voltage divided by the current equals the resistance, as shown in this equation:

Resistance = $\frac{\text{Volts}}{\text{Amps}}$

or

$R = \frac{V}{I}$

Value of resistor (Ohms)	0.5	1.0	
Reading on Voltmeter (V)		4	4
Reading on Ammeter (I)	2		2

- 9

Use the equation above to complete the table at the right.
- 10

Draw the following circuits.

a

A series circuit with a power pack using 8 volts, a 6 watt globe and two 4 ohm resistors.

b

A parallel circuit with a power pack using 8 volts, a 6 watt globe, an ammeter and three 4 ohm resistors.

Fossils

Fossils are formed from the hard parts of a shell or a skeleton. With plant fossils, it is usually the leaves and the woody parts that are found. Fossils can form as impressions, moulds and casts. They can be found in amber (a plant resin), ice and sedimentary rocks such as coal, shale and sandstone. Fossils provide important evidence of past life, and relate the evolution of organisms.

- 1 Explain why fossils found in rocks are usually found only in sedimentary rocks.

The process of fossilisation takes time. When an organism dies and falls into soft sediments, such as those found in lakes and ocean floors, the soft parts usually decay very quickly, leaving only the hard parts. The processes of fossilisation are: mineralisation, replacement, re-crystallisation. Moulds and casts can be formed.

- 2 Write these words in the correct order to summarise the process of fossilisation: mineralisation, death, sediment burial, replacement, decay, compaction.

The remains of the organism become compacted and mineralised, and eventually the layer turns to rock. The rock layer eventually becomes exposed.

- 3 The largest herbivorous dinosaur in Australia was recently discovered in Queensland. The fossil was embedded in sandstone. Describe how this dinosaur was fossilised.

Fossils are found in rock layers. The oldest fossils are in the deepest rock layer while the youngest fossils are found in the layers nearest to the surface. This is the basis of the Law of Superposition.

- 4 Explain why more advanced fossilised organisms are found in younger rocks.

- 5 Use the rock correlation diagram at the right to answer the following questions.

a Correlate the rock layers according to the fossils.

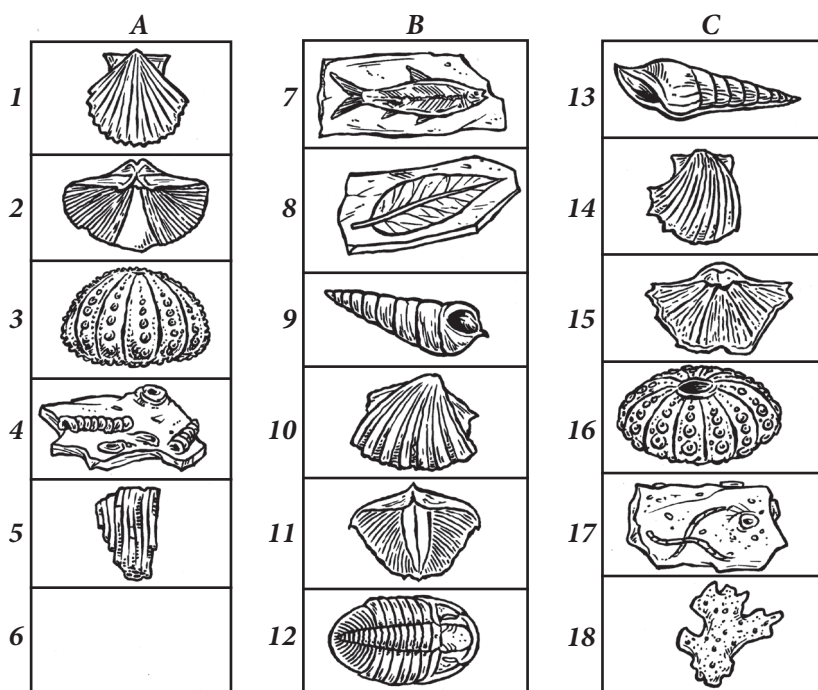
b State which fossil is the oldest and which is the youngest.

c Describe the environment in which the leaf fossil (8) formed.

d State which fossil lived in a warm marine environment.

e Describe the environment in which the snails and worms lived.

Localities



Numbers 1, 10 and 14 are bivalves. 2, 11 and 15 are brachiopods. 9 and 13 are snails. 3 and 16 are echinoderms. 4 and 17 are crinoids and worms. 5 and 18 are corals. 7 is a fish. 8 is a leaf and 12 is a trilobite.

Hormones and nerves

Hormones are chemical messengers made by endocrine glands. These glands secrete hormones directly into the circulatory system. Each hormone has a specific chemical function and is received by specific target cells causing a specific bodily response. The pituitary gland is called the master gland. It is about the size of a pea, and found at the base of the brain. The pituitary secretes stimulating hormones such as luteinizing hormone (LH), follicle stimulating hormone (FSH), adrenocorticotrophic hormone (ACTH), and thyroid stimulating hormone (TSH). The pituitary produces growth hormone and antidiuretic hormone (ADH).

Luteinizing hormone triggers ovulation in females, and in males triggers the production of testosterone and the manufacture of sperm in the testes. Follicle stimulating hormone stimulates the early maturation of ovarian follicles, and in males the production of mature sperm. ACTH targets the adrenal glands that sit on top of the kidneys to produce steroids. ACTH is also necessary for the production of the sex hormones oestrogen and testosterone. Progesterone is essential for the maintenance of pregnancy and the proper functioning of the menstrual cycle. Progesterone is found in both males and females as it assists with raising body temperature.

1 Mark each statement as true or false. Then correct the false statements so that they are true.

- a** ACTH is produced by the pituitary gland. _____
- b** FSH is important for the maturation of ova and sperm. _____
- c** The master gland is the pituitary gland. _____
- d** Progesterone is needed to lower body temperature. _____
- e** Testosterone is needed to produce secondary sex characteristics in males. _____

TSH stimulates the thyroid gland in your neck, which releases thyroxine into the blood stream to control metabolism. There are several target organs for thyroxine, but the chief target organ is the liver. Iodine is essential for the production of thyroxine and for ensuring proper neural brain development and connections. The pancreas, under your stomach, produces insulin and controls blood glucose (sugar) levels. Glucagon is another hormone produced by the pancreas and it raises the glucose level in the body, particularly during exercise.

2 Use the information above to label the glands of the body on the diagram at the right.

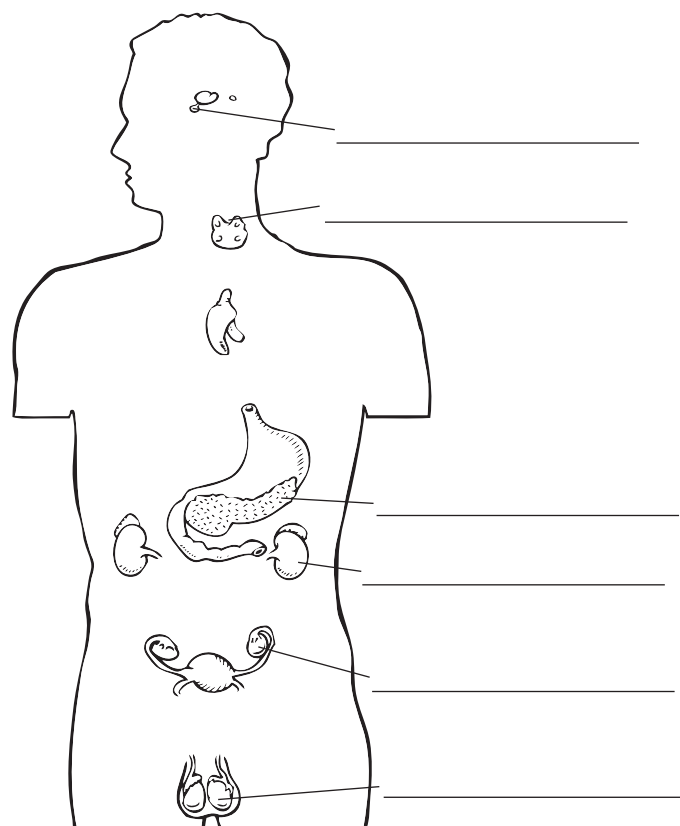
3 Which condition is the result of insufficient insulin in the body?

The more anti-diuretic hormone (ADH) released, the more water absorbed by the body, and the less urine produced. ADH controls the amount of water and salt in our bodies. Our kidneys reabsorb water.

Growth hormone targets most cells in the body, and is responsible for the growth of the body.

Adrenalin, the 'fight or flight' hormone, is produced by the adrenal glands. Adrenalin increases heart rate, raises blood pressure and causes the release of sugar into the blood from the liver. Adrenalin targets the muscles, heart and liver.

4 What are target cells?



5 Complete the table below.

GLAND	LOCATION IN BODY	HORMONE	NAME OF TARGET ORGAN	FUNCTION
		Insulin		
	Base of brain			Production of sex hormones
			Adrenals	
		ADH		
			Testes	
	Ovaries			
Thyroid				
		GH		

6 Complete the information below using these words: axon, neurones, dendrites, myelin, neurotransmitters. (Some words are used more than once.)

The nervous system is made up of _____ and makes and transmits electrical signals. Electrical impulses travel down the _____. Coming out from the neurone are small branches called _____. They receive messages from other neurones. The _____ branch out to stimulate a muscle or to stimulate the secretion of a hormone. The axon terminal of a neurone releases _____ (chemicals) in response to an impulse across a small gap. Most axons are covered in a _____ sheath.

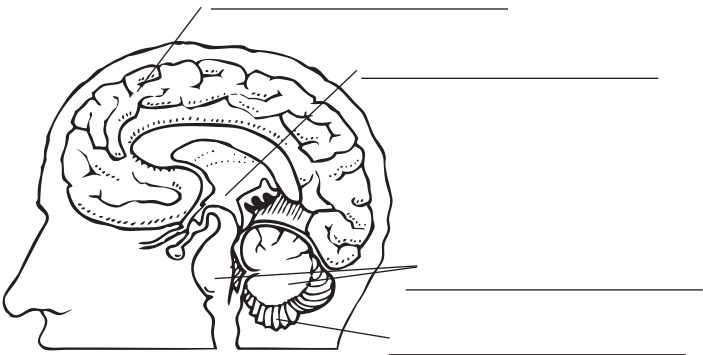
The nervous system is divided into several components.

The central nervous system (CNS) is the brain and the spinal cord. The brain is composed of a cerebrum that carries out intellectual functions, voluntary movements, sense interpretation, speech and personality. The cerebellum coordinates movement, and controls repetitive movement, balance and posture. The brainstem regulates involuntary movements such as heart rate and breathing. The cerebellum and brainstem make up the hindbrain.

Receptors are specialised nerve cells that detect pain, temperature, light, sound and smell. Sensory nerves transmit messages to the CNS from the receptors. Effectors are muscles or glands that are stimulated by motor neurones, producing an electrical or a hormonal response. Motor neurones transmit messages from the CNS to the effector cells.

Rapid reflex action such as blinking or removing your hand from something hot is called a reflex pathway. The receptors transmit the message to the sensory neurone, which in turn stimulates the motor neurone in your spinal cord to contract, via effectors, the muscle in your arm or eye. A subsidiary message is sent to the CNS to inform you where you have hurt yourself. Reflex arcs such as the knee reflex are involuntary and do not send subsidiary messages to the CNS.

- 7 In your workbook, draw and label a picture of a reflex pathway of a bare-foot person standing on a nail.
- 8 Label the components of the brain in the diagram at the right using these words: cerebellum, cerebrum, hindbrain and midbrain.
- 9 Draw a mind map to show the components of the central nervous system and how these components relate to each other.



Indicators and pH

Indicators are used to find out if a substance is acidic or basic. They are dyes that can be derived from various plants such as nasturtiums, red cabbages and hibiscus flowers. In the year 1300, Villanova discovered that lichens can be used to make litmus paper. Blue litmus paper turns red when dipped in acid and red litmus paper turns blue when it is dipped into a base. Not all chemicals react with litmus paper; for example, chlorine gas bleaches litmus paper white.

Indicators such as universal indicator paper contain many dyes that change colour gradually over a wide pH range. Other indicators such as bromophenol blue, phenolphthalein and methyl red are very specific in their ranges. The table at the right shows some indicators and their pH ranges. Indicators are used for testing soils, pool water, bodily fluids and other chemical substances.

pH is the logarithmic scale of the hydrogen ions (H^+) in a solution. A pH of 0 is strongly acidic, and a pH of 14 is strongly basic (OH^-). A neutral pH is 7.

The pH level varies in bodily functions. The human stomach has a pH of 1 because it produces hydrochloric acid. Other parts of the digestive system have a higher pH. Human blood has a pH of 7.3–7.5. If these levels are exceeded in either direction, the body cannot process properly the ingested foods and gases.

INDICATOR	PH RANGE
Methyl violet	0.0–2.0
Methyl yellow	2.9–4.0
Congo red	3.0–5.0
Methyl red	4.4–6.2
Azolitmin	4.5–8.3
Bromocresol purple	5.2–6.8
Bromothymol blue	6.0–7.6
Phenol red	6.8–8.4
Cresol red	7.2–8.8
Thymol blue	8.0–9.6
Phenolphthalein	8.2–10.0
Alizarine yellow	10.2–12.0

1 Which three indicators in the table could you use to show a pH of 8.7?

2 Many soils have a pH of 7–8.5.

a Is soil generally basic or acidic? _____

b Which indicator would be best for testing soil? Give a reason for your answer.

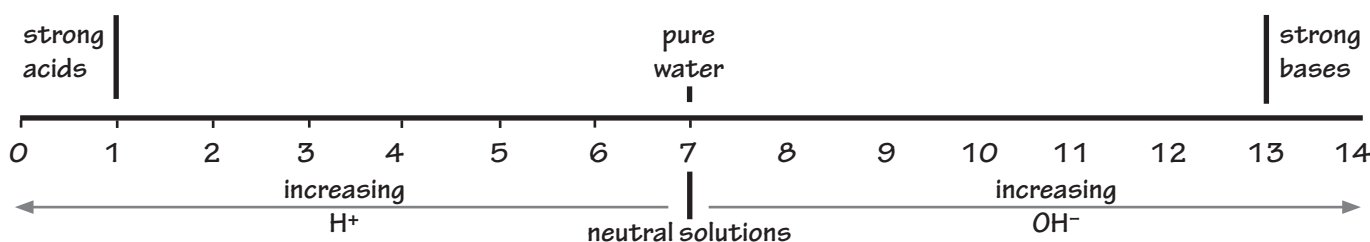
3 a Which indicator in the table has the widest range? _____

b Describe a situation in which using this indicator would be an advantage.

4 a Which indicator in the table has the smallest range? _____

b Describe a situation in which using this indicator would be an advantage.

5 Write these substances in the correct place on the pH scale below: lemon pH 2.0, blood pH 7.4, sea water pH 7.3, sodium hydroxide pH 14.0, tomato pH 4.5, stomach HCl pH 1.0, mango pH 3.0, fresh water pH 8.0.



Joules, watts, amps and volts

Joules are a measure of energy. You use energy every day and you need to maintain a level of energy to keep you functioning, even while you sleep. Every action you do expends energy.

James Prescott Joule was an English physicist, born in 1818 in Salford, Lancashire. Joule studied the nature of heat, and discovered its relationship to mechanical work. He also stated that heat is a form of energy. This led to the theory of conservation of energy. The joule is the international scientific unit for measuring energy.

- 1** What do you think the theory of conservation of energy is?
-

Think of a joule as the energy used when a 100 g mobile phone is lifted through a height of 1 metre and held against the force of gravity, which is almost 10 Newtons. One joule is defined as the amount of energy exerted when a force of one Newton is applied over a displacement of one metre. A joule is also the equivalent of one watt of power being radiated for one second. This means that the energy in food can be related to the amount of work that can be done from specific food items. Food energy is commonly measured in kilojoules (kJ).

- 2** A chocolate biscuit has 300 kilojoules of energy. How many times must a 100 kg person walk up a 5 m high stair case to use up this energy? (Assume the force of gravity = 10 Newtons)
-

Consider a 9 volt battery with a current of 1 amp. This means that 9 joules per second (or 9 watts) are being used. When one joule is used per second, the unit of power is called a watt, after James Watt. One watt of power is equivalent to the rate at which energy is produced by a candle. Think of a watt as a quantity of electrical energy flowing per second. The amount of power being used is the rate at which energy is expended, or Power = Volts x Amps, which can be written as $P = VI$.

This can also be expressed as:

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken in seconds}}$$

$$\text{or } P = \frac{F \times d}{t} \quad \text{where } F \text{ is the force, } d \text{ is the distance and } t \text{ is time.}$$

$$P = I^2 R \quad \text{where } I \text{ is the amps and } R \text{ is the resistance in ohms}$$

- 3** Use the above formulae to calculate the answers.

- a** How many joules are used when 6 amps flow through a 2 ohm resistor in 1 second? _____
- b** Calculate the amount of energy when 50 Newtons are expended in 5 seconds. _____
- c** How many amps are used if 10 volts are used in 1 second and the power used is 125 joules? _____

The volt is named after the Italian physicist Count Alessandro Volta who lived between 1745 and 1827. Volta invented the first chemical battery in 1800. The ampere is named after André-Marie Ampère, a French physicist who lived between 1775 and 1836. The ampere or amp is the amount of electrical charge per second. Electrically, when the potential difference (or one volt) is applied to a resistive load, and carrying a current of one Ampere, then one watt of energy is radiated.

- 4** Explain the difference between a watt and a joule.
-

- 5** State whether or not the following produce energy in joules. Give reasons for your answers.

- a** A Bunsen burner burning gas _____
- b** A person lifting a bag _____
- c** Hitting a cricket ball _____
- d** A person leaning on a table _____
- e** A motorcycle speeding down the road _____

Motion

Speed is the word we use to describe how fast an object is moving. Speed is calculated by the distance travelled divided by time. In science, it is measured in metres per second. The equation for speed is:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Velocity is not the same as speed. Velocity does describe how fast an object is moving, but it also describes the direction in which the object is moving. Therefore a change in the direction of an object is a change in velocity, but not necessarily a change in speed. Velocity is calculated the same way speed is, but the units are usually in metres per second plus direction, e.g. 30 ms⁻¹ east.

Acceleration is the change in the velocity of a moving object, and can refer to an increase in velocity or a decrease (negative acceleration). The equation for acceleration is:

$$\text{acceleration} = \frac{\text{final velocity} - \text{initial velocity}}{\text{time taken to change velocity}} \quad \text{OR} \quad \frac{V - u}{t}$$

The units are expressed in metres per second per second or ms⁻².

Acceleration is directly related to Newton's Law: Force = mass × acceleration.

Note: For the following questions, convert all distances to metres and all times to seconds.

1 Calculate the following.

a What is the average speed in km/h of a motorbike when it travels 20 km in 15 minutes?

b A motorboat travels 72 m/s in an easterly direction for 4 km. What is the time taken?

c You have swum for 15 minutes at a speed of 0.5 m/s. What is the distance you have travelled?

Motion is often represented with graphs. The steeper the gradient, the faster the acceleration or deceleration. A horizontal line indicates that the object is moving at a constant rate.

2 Label the following on the graph at the right.

a axis showing velocity

b axis showing time

3 Write the points between which each of the following is occurring on the graph.

a constant velocity _____

b fastest acceleration _____

c fastest deceleration
(negative acceleration) _____

d slowest acceleration _____

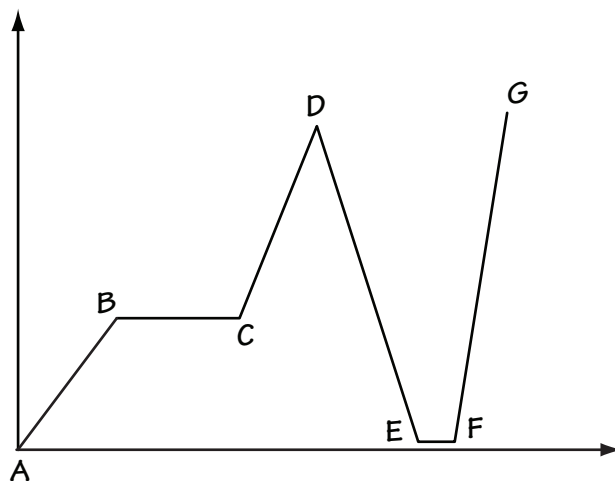
4 Referring to the graph on the next page, write the velocity of the truck at each given time.

a 8 seconds _____

b 1.5 seconds _____

c 6.25 seconds _____

Velocity-time graph



d 10 seconds _____

5 Could you calculate the distance the truck travelled in the 12 seconds shown in the graph? Give reasons for your answer.

6 a A motorcyclist is stationary at a red traffic light. The light turns green and the motorcycle reaches 60 km/h in 4 seconds. Calculate the acceleration.

b What is the force of the motorcycle if it has a mass of 150 kg?

7 A deflating balloon travels 8 metres through the air in 5 seconds. What is the average speed of the balloon?

There are two other equations used in calculating motion components. These are:

$v = u + at$

Where v = final velocity, u = initial velocity, a = acceleration and t = time taken to accelerate.

$d = ut + \frac{1}{2}at^2$

Where d = distance, a = acceleration, t = time taken to accelerate and u = initial velocity.

Acceleration can be the acceleration due to gravity, which is 9.8 ms^{-2} .

8 A motorbike was travelling at 35 m/s. It overtook a truck by accelerating at 4 ms^{-2} for 4 seconds. How far did the motorbike travel in 4 seconds?

9 Calculate the final velocity of an object that was travelling at 6 m/s with an acceleration of 2 ms^{-2} for 4 seconds.

10 A cricket player hits a ball with a mass of 90 g and it is found that the acceleration of the ball is 25 ms^{-2} . What was the driving force?

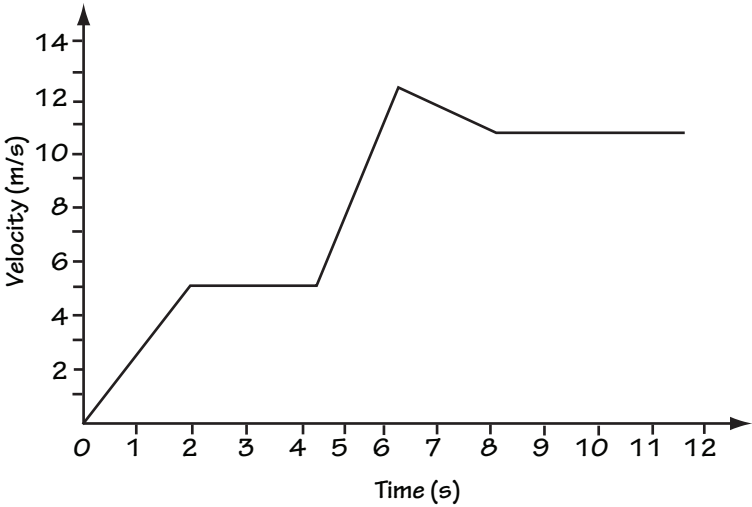
11 A car has a mass of 1500 kg and when the brakes are applied there is a force of 12 000 N.

a What is the deceleration of the vehicle? _____

b How long will it take the vehicle to come to a stop from a speed of 30 m/s? _____

c From your answer to part b, how far will the vehicle travel before coming to a stop? _____

Truck velocity over time



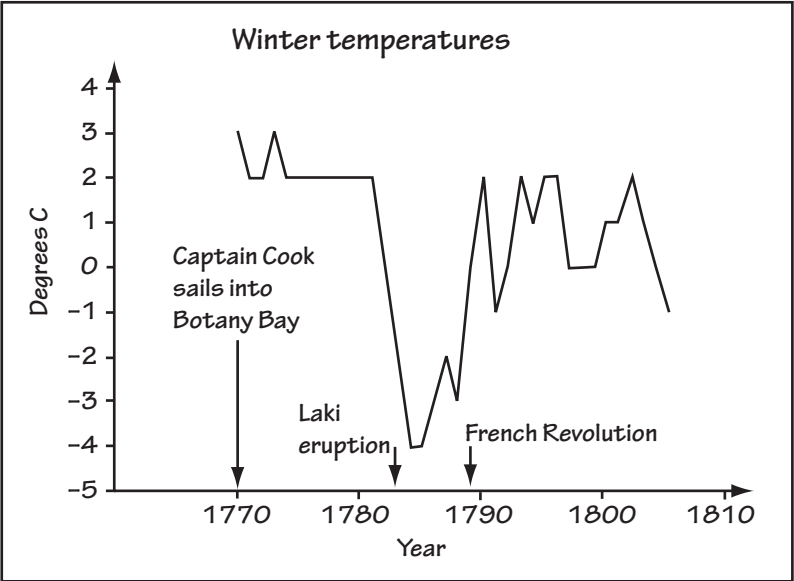
Natural events

Volcanoes have provided the Earth with its atmosphere, water vapour and materials to make productive soils. Volcanoes also alter climatic conditions, because the volume of ash that is put into the atmosphere reduces the amount of the Sun's radiation. The Mt Pinatubo eruption in the Philippines caused the temperature to fall one degree, inducing heavy rainfall. In Iceland, the Laki fissure volcanic eruption in 1783 caused Europe to have a prolonged cold spell for several years. Like cyclones, volcanoes can cause millions of dollars' worth of damage.

1 Referring to the graph, describe the problems associated with the temperatures and agricultural production in Europe, Scandinavia and Iceland.

Temperature effects

Agricultural effects



2 From the graph, suggest how the Laki eruption might have triggered the French Revolution.

Cyclones are the same as typhoons and hurricanes. They are generated out at sea due to temperature and pressure differences. The rain that is generated in a cyclone dilutes the upper layer of the sea and can affect the growth of corals. The rain also causes flooding and can kill animals that live in burrows as well as causing major erosion. The wind can cause damage to coral reefs and land vegetation as well as having an emotional effect on people. With climate change, cyclones will become more frequent and affect areas not usually prone to them.

3 Tropical Cyclone Larry crossed the coast near Innisfail in Far North Queensland in March 2006. Use these terms in sentences to show the correct time order sequence of the events associated with Cyclone Larry: strong winds, loss of agricultural produce, low pressure develops offshore, torrential rain, erosion, destruction of houses, human emotional distress, appointment of General Cosgrove to administer the reconstruction, habitat loss.

4 In your workbook, write three paragraphs explaining how cyclones affect the biosphere, the lithosphere and the hydrosphere. Use the definitions below to help you.

- Biosphere – the ecosystem comprising the entire Earth and all living organisms that inhabit it.
- Lithosphere – The outer part of the Earth, consisting of the crust and upper mantle.
- Hydrosphere – All the Earth's water; oceans, lakes, rivers, groundwater, snow cover, ice, and water vapour in the atmosphere.

Newton's Laws

Basically, Newton's Laws are intuitive – see that book over there? You'd get a shock if it moved all by itself. You expect things to stay in motion or motionless, unless acted upon by a force.

According to Newton's First Law, an object will only change the way it is accelerating or moving if it is acted on by an unbalanced force, or not cancelled out by an equal force in the opposite direction. This means that objects will remain moving or stationary with the velocity they already have. In the absence of an unbalanced force, an object in motion will maintain its state of motion. This is often called the Law of Inertia. Inertia resists changes in its velocity.

1 a Loose objects in a car move forward when the car stops suddenly. That is, they continue to move forward as the car stops. Explain in terms of forces why this happens.

b Why are seat belts necessary in cars? Include an example of a specific situation in your answer.

2 The aisle in a bus is slippery. Predict the outcomes for a bus that is full of people if the bus:

a starts moving suddenly. _____

b suddenly turns a corner. _____

c accelerates rapidly from a stop sign. _____

Newton's Second Law states that the acceleration of an object is dependent upon the unbalanced force acting on the mass of the object. That is, a large force = large acceleration, a small force = small acceleration. This concept can be expressed as:

acceleration = Force ÷ mass or $a = F \div m$

force = mass × acceleration or $F = ma$

Study the examples shown below.

Example a

Force = mass 10 kg × acceleration 4 ms⁻² = 40 Newtons

Example b

Acceleration = $\frac{\text{Force 20 Newtons}}{\text{mass 10 kg}} = 2 \text{ ms}^{-2}$

Example c

Mass = $\frac{\text{Force 40 Newtons}}{\text{acceleration 4 ms}^{-2}} = 10 \text{ kg}$

3 a Calculate the missing values to complete the table at the right.

b A trolley is acted upon by a force of 60 Newtons and its mass is 15 kilograms. Calculate its acceleration.

c Sue's bike has a mass of 150 kg. If the acceleration of the bike is 10 ms^{-2} , what is the force?

d What is the mass of an object falling with an acceleration of 9.8 ms^{-2} and a force of 833 Newtons?

FORCE (N)	MASS (kg)	ACCELERATION (ms^{-2})
100	5	
	4	20
	6	15
20		5
36	6	

When you see a car move forwards, look carefully and you will see that the wheels are turning backwards. This is Newton's Third Law of Motion in action.

Formally stated, Newton's Third Law is: *For every action, there is an equal and opposite action.*

This means that in every interaction, there is a pair of forces acting on the two interacting objects. The size of the force on the first object equals the size of the force on the second object, but is in the opposite direction.

4 Identify the action and reaction in each of the following interactions.

a A book resting on a table

b A rocket accelerating in space

c A ship floating on water

d A gun firing a bullet

e A person leaning on a fence

5 Friction is very important when considering forces. Explain why.

6 In the space below, draw a picture of a skateboarder travelling down a footpath and colliding with a light pole. Label it to identify the forces that are involved. Then write which of Newton's Laws relates to the collision.

Non-renewable resources

Non-renewable resources are the fuels that cannot be replaced; for example, coal and oil, which were formed over millions of years. These fuels cannot be replaced in our lifetimes, so should be conserved carefully.

The original source of energy for Earth is the Sun. The energy comes to Earth in the form of electromagnetic waves. Plants convert sunlight into carbon-based carbohydrates. In turn, the plants provide energy to the animals that feed on them.

Coal is formed when layers of buried plant material are compressed and altered. Oil is formed from large amounts of small marine organisms that have been buried for millions of years.

We burn oil in our cars as petrol and in planes as kerosene. Industry burns coal to produce electricity. Heat from coal is also used to produce metals, and coal is used to remove the oxygen from the metal ore.

1 What is the original source of the energy on Earth? _____

2 Circle the fossil fuels in the box below.

+ natural gas	+ hydroelectric energy	+ oil	+ petrol
+ biodiesel	+ geothermal	+ nuclear energy	+ coke
+ solar energy	+ coal	+ aviation fuel	+ wood

3 Write five steps to show how the Sun's energy is turned into coal.

_____ → _____ → _____ → _____ → _____

4 a List two uses of oil. _____

b List two uses of coal. _____

5 Describe how the burning of fossil fuels has contributed to the Greenhouse Effect.

6 Use these words to complete the sentences: coal, oil, fossil fuel, energy, combustion, electricity.

a Plants need the Sun's radiant _____.

b Coal-fired power stations generate _____.

c Petroleum or rock oil is a _____.

d Long buried marine micro-organisms form _____.

e Plants fall into swamps producing _____.

f Carbon dioxide is released during _____.

The metals that people mine come from either metal oxide ores or metal sulfide ores. Precious metals such as gold and platinum are mined, then heated and chemically treated to remove the impurities in the crushed ore. Both metals are usually 99% pure after this process.

7 Complete the following equations.

a iron III oxide + carbon → iron + _____ **b** nickel sulfide + oxygen → nickel + _____

Uranium oxide is a radioactive metal that is chemically treated to extract the uranium. Australia has large uranium deposits. Most of the uranium is exported to countries with nuclear power plants. One problem with nuclear power plants is that the by-products can be used for manufacturing nuclear weapons. Another problem is that the waste products need to be stored for thousands of years.

8 In your workbook, write five brief arguments for uranium mining and five arguments against.

9 Eventually all the metals and fossil fuels that we use will be depleted. Write a short journalistic article about the recycling methods and alternatives that people could use.

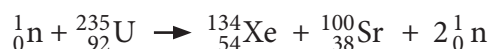
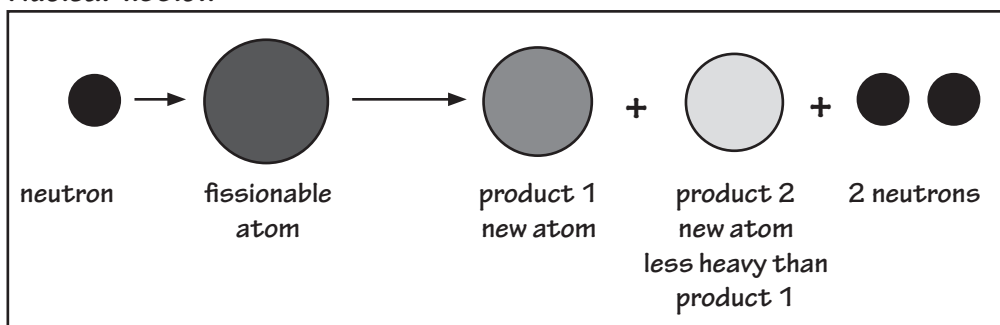
Nuclear fission and fusion

Fission

Nuclear fission is valuable as it is used in atomic power plants, warships and nuclear weapons. The particles in an atomic nucleus are held together by a very strong force. When a neutron is fired at the nucleus, the atom splits into two or more new products, breaking the atom down and releasing energy for people to use. This is called fission or splitting. Not all nuclei undergo fission.

As a result of nuclear fission, a chain reaction occurs, generating heat and radiation. If this chain reaction is not controlled, a nuclear explosion occurs.

Nuclear fission

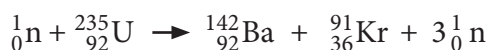


- 1 a** Write a sentence to describe what is occurring in the above equation. Use these words: uranium, xenon, strontium, neutron.

- b** State the meaning of the superscript and subscript numbers in the equation above.

Superscript: _____

Subscript: _____



- 2** Write a sentence to describe what is occurring in the above equation. Use these words: uranium, barium, krypton, neutron.

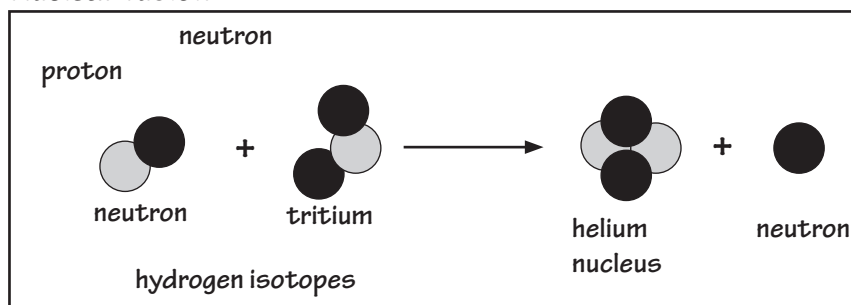
Fusion

Nuclear fusion is the combining of light elements such as hydrogen to form a heavier element such as helium. When the nuclear particles collide, they release energy.

There are two main technological reasons why fusion is not used for generating electricity today:

- » How to force the two nuclei together – this requires a great deal of energy because of the mutual repulsion that needs to be overcome.
- » How to contain the plasma or gases of the reaction.

Nuclear fusion



A typical fusion reaction is:



- 3** Write a word equation for the above equation. Use these words: deuterium, tritium, helium, neutron

Our Sun and other stars undergo fusion reactions. The reactions continue until iron nuclei are formed. At this point, the star usually explodes.

Fission reactors split an atom, and release heat to turn water into steam. This in turn runs a generator to make electricity. Experimental nuclear fusion reactors work using high speed circular accelerators to fuse the elements together.

- 4 a** Is it environmentally safer to run nuclear-powered electricity stations or coal-powered electricity stations? Explain your answer.

- b** In your workbook, construct a table to show the advantages and disadvantages of nuclear-powered electricity generation and coal-powered electricity generation.

Elements such as uranium and plutonium are radioactive, that is they release particles spontaneously. These particles can be (α) alpha particles (helium nuclei), (β) beta particles (high-speed electrons) and (γ) gamma rays (high energy electro-magnetic radiation). Beta particle decay occurs when a neutron is changed into a proton within the nucleus.

Isotopes are elements with the same number of protons but have a different number of neutrons. Isotopes can be radioactive or stable. In alpha decay, the mass number ($222 \rightarrow 218$) and atomic number ($86 \rightarrow 84$) are reduced by 4 and 2 respectively. In beta decay the mass number of the product stays the same but the atomic number increases by 1 as the neutron is converted into a proton.

- 5** Complete the following equations by noting the numbers of protons and neutrons that are changed and then state if they are alpha or beta decay reactions.



- 6** Describe the similarities and differences between nuclear fission and nuclear fusion.

Periodic Table

Dmitri Mendeleev devised the Periodic Table, and it was first published in 1869. The table is divided into eight columns called **groups** and rows called **periods**. The groups are numbered from 1 to 8, and these numbers refer to the number of electrons in the outermost shell of each element. Each outer electron shell can hold a fixed number of electrons. Group 1, which is very reactive, has 1 electron. Group 7 has seven electrons and is also very reactive. Group 8 has eight electrons and the elements do not react readily, if at all. Each period is also numbered and tells you how many electron shells there are in each row.

The electronic configuration of an element is written by the number of electrons in each shell, e.g. 2.8.3. The configuration tells you that the element is in Group 3 from its outermost shell with 3 electrons and that it is in the third row of the Periodic Table because there are 3 shells of electrons. This identifies the element as aluminium.

The name and atomic number of an atom of the first 20 elements are listed at the right.

1 Complete the table by identifying the group, row and element for each electronic configuration.

Electronic configuration	2.8.8	2.1	2.4	2.8.7	2.8.8.1
Group					
Row					
Element					

- hydrogen 1
- helium 2
- lithium 3
- beryllium 4
- boron 5
- carbon 6
- nitrogen 7
- oxygen 8
- fluorine 9
- neon 10
- sodium 11
- magnesium 12
- aluminium 13
- silicon 14
- phosphorus 15
- sulfur 16
- chlorine 17
- argon 18
- potassium 19
- calcium 20

In an electrically balanced atom, the number of electrons must equal the number of protons in the nucleus. The number of protons in the atom is equal to its atomic number. The higher the atomic number, the further it is down the Periodic Table.

2 Given the atomic number, write the electron configuration of each element; e.g. atomic number 16, electron configuration 2.8.6.

- a** Atomic number 3, electron configuration_____
- c** Atomic number 12, electron configuration_____
- b** Atomic number 20, electron configuration_____
- d** Atomic number 5, electron configuration_____

Protons are found in the nucleus and are electrically positive. Neutrons also are found in the nucleus but have no electrical charge. Protons and neutrons give the element its mass number. Electrons are found in ‘shells’ around the nucleus. The first shell can have only 2 electrons, while the second, third and fourth shells will fill up progressively until they have 8 electrons. Having an outer shell with 8 electrons provides the atom with a noble or inert gas configuration, which makes it stable.

3 In the box below right, draw the atom with the electronic configuration 2.8.4. Make sure you include the correct number of protons and neutrons, and the electron shells with the correct number of electrons in each shell.

The properties of the elements across the Periodic Table, with the exception of hydrogen and helium, from left to right are:	The properties of the elements of the Periodic Table from top to bottom are:
• a decrease in metallic character • a decrease in the atomic radius • bonding goes from ionic to polar covalent to covalent • metallic to semi-metals to non-metals.	• an increase in metallic character • a decrease in the atomic radius • bonding more ionic.

4 From the information above, state the chemical properties of these elements.

- a** oxygen
- b** lithium
- c** silicon

Quasars and stars

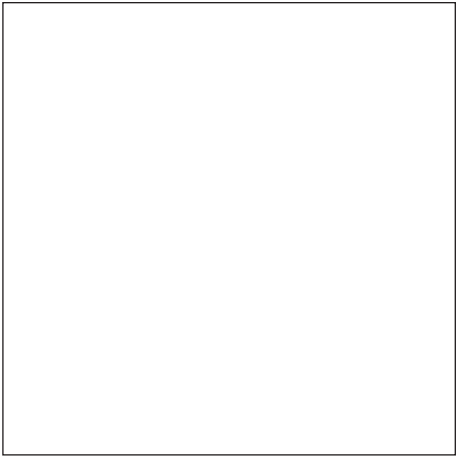
Quasars are the brightest and most distant objects in the known universe. They burn with the energy of a trillion suns and are a strong source of radio waves, which are detected by radio telescopes. Quasars are not true stars. The word ‘quasar’ means *quasi-stellar radio source*. Quasars have a very high red shift, which means they are moving away from the observer very quickly. The nearest quasar to Earth is 10 billion light years away.

1 a What are the advantages of using radio telescopes rather than optical telescopes to detect quasars?

b What conclusion does the red shift of the quasars suggest about our universe?

Nebulae are gigantic gas clouds and are the basic building blocks of the universe. They contain the elements from which stars and solar systems are built. Nebulae are composed of about 90% hydrogen, 10% helium and 0.1% heavy elements such as carbon, nitrogen, magnesium, potassium, calcium and iron.

2 In the box at the right, draw and label a pie chart to show the elements contained in a nebula.



A **pulsar** is formed when a massive star collapses and exhausts its supply of fuel. It blasts out in a giant explosion known as a supernova. If the mass of the core of the collapsing star is 1.4 times the mass of the star itself, it forms a neutron star. If it is less than this, it becomes a white dwarf. If the mass of the core is greater, a black hole results.

3 a Describe the differences between a neutron star and a white dwarf.

b Outline the life cycle of a pulsar.

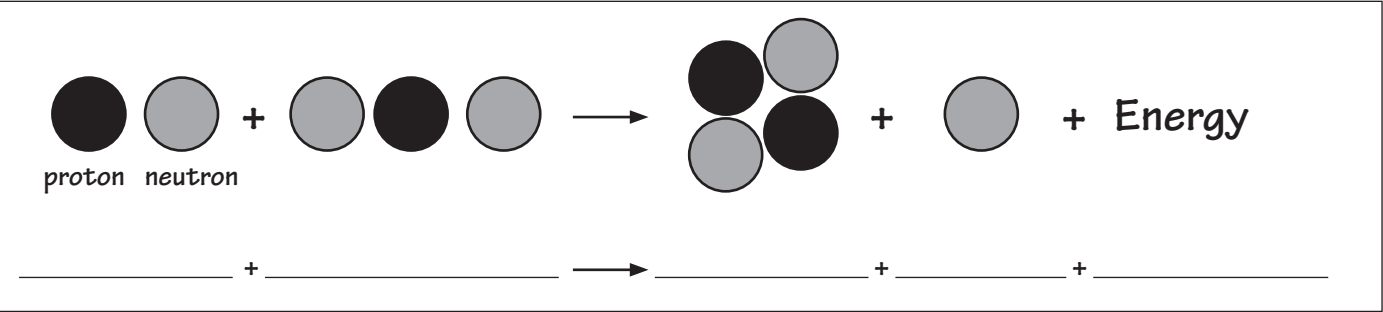
c Name the object that has a massive gravitational force and destroys matter.

A **star** is like a gigantic nuclear furnace, where hydrogen is converted into helium plus energy. This process requires the two isotopes of hydrogen, deuterium (one proton and one neutron) and tritium (one proton and two neutrons). When the hydrogen isotopes are used up, the helium is converted to oxygen and carbon. If the star is large enough, it will convert these into neon, sodium, magnesium, sulfur and silicon. This will continue until all the elements are converted to calcium, nickel, chromium, copper and others until iron is formed. When the core of the star is primarily iron, it implodes on itself.

d Draw three or more diagrams in the space below to represent the above information about stars.

4 Complete the word equation for nuclear fusion.

Fusion reaction



A **star** that maintains a stable nuclear reaction is called a main sequence star. These stars only fluctuate slightly in terms of temperature and luminosity, and this is dependent on its mass. Stars like our Sun will burn for several billion years. On the other hand, massive stars will only burn for a few hundred million years and will turn into blue giants.

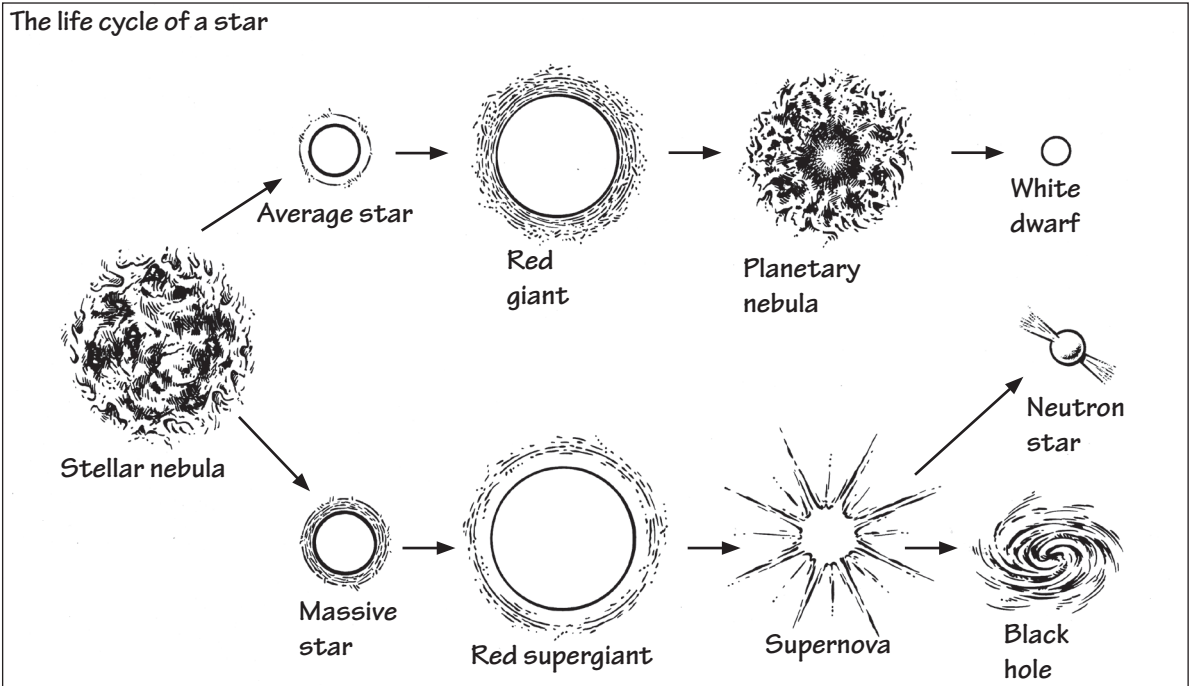
All stars are born in a high-density nebula. As the condensing matter heats up and nuclear reactions start, hydrogen fuses to form helium. At this point they are generally yellow stars like our own Sun. As energy is released, the outer contraction stops forming a main sequence star. Due to the nuclear fusion reactions requiring more fuel, the helium core starts to contract. The core then begins to fuse to form carbon, and the outer layer shines less brightly and begins to expand forming a red giant. As the gases drift away, the core becomes a white dwarf and begins to dim. When it stops shining, the core develops into a black dwarf.

When a protostar fails to reach its critical nuclear mass, it becomes a brown dwarf. Brown dwarfs are giant gas balls that are too large to be a planet but have insufficient mass to be a star. They emit no light or heat.

5 Complete the table by listing two features of each star.

FEATURES	YELLOW STAR	RED GIANT	BLACK DWARF

The life cycle of a star



6 Imagine you are the Sun, an average star. In your workbook, write about your birth, development and old age. Use the information from the diagram above to help you.

Reflection and refraction

Light travels in straight lines, but it can be bounced (reflected) back on an angle, depending on the angle of the surface that it strikes. The ray of light that strikes a surface is called the incident ray. The ray that is reflected from the surface is called the reflected ray. If a ray strikes the surface at 90°, it is called a normal ray.

The Law of Reflection states that when a ray of light reflects off a surface, the angle of incidence is equal to the angle of reflection.

1 Complete the table at the right.

If you look in a plane (flat) mirror, the reflection you see will seem to be the same distance away. That is, if you are 30 cm away from the mirror, the image will seem to be 30 cm behind the mirror.

2 Complete the diagram at the right by drawing in the incident ray, the normal ray, the reflected ray and the virtual image.

There are two types of curved mirrors: concave and convex. Concave mirrors work in much the same way as plane mirrors, but have a focal point (F) and a principal axis (normal). Curved mirrors are important for traffic control, headlights and microscopes.

There are several lens types, which may be converging or diverging, and they have different functions. The human eye is a biconvex lens, as are the lenses used in magnifying glasses.

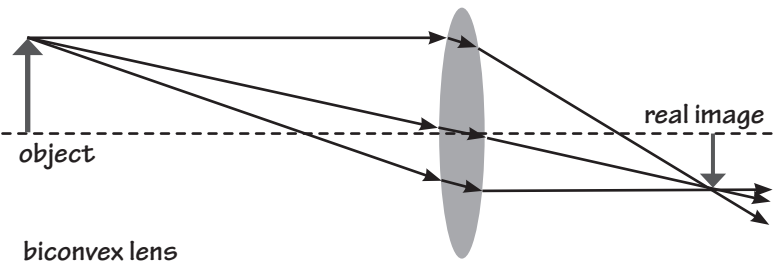
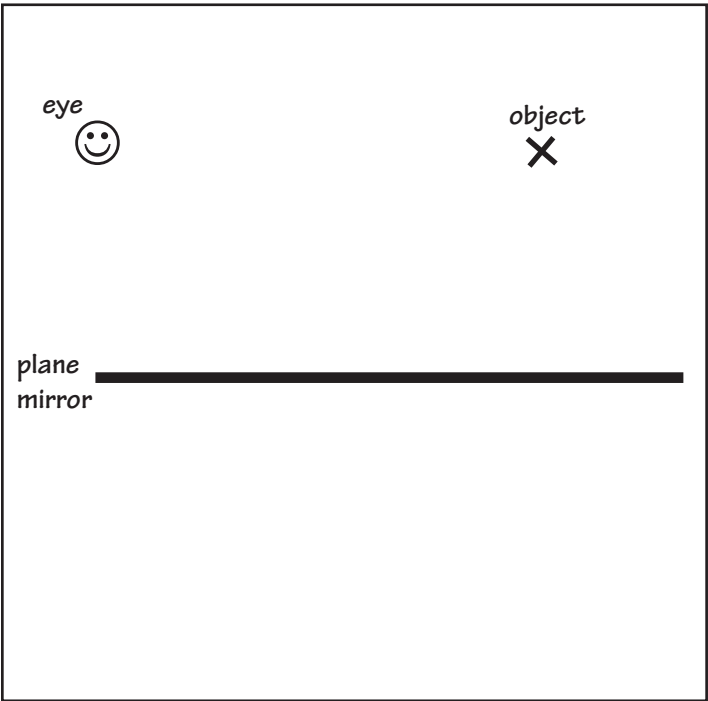
3 Is a biconvex lens converging or diverging?

4 Examine the diagram below right.

a Compare the object and the real image.
What do you notice?

b What does this suggest about the human eye and the human brain?

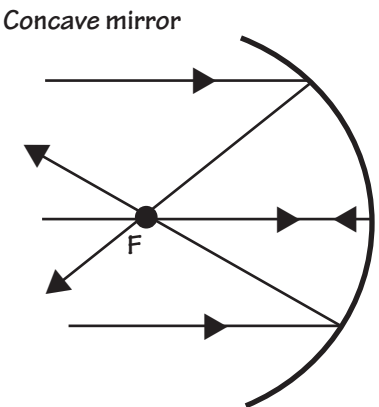
ANGLE OF INCIDENCE	ANGLE OF REFLECTION
3	
5	5
8	
	11



Concave mirrors

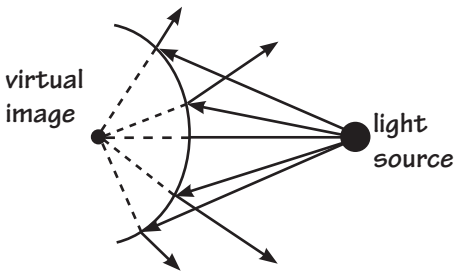
5 Match the points below to the rays on the diagram of the concave mirror by labelling the diagram with a, b and c.

- a** Any incident ray travelling parallel to the principal axis on the way to the mirror will pass through the focal point upon reflection.
- b** A ray that passes through the focal point to the mirror will reflect and travel parallel to the principal axis.
- c** The ray that travels parallel to the principal axis to the mirror will reflect and travel through the focal point.



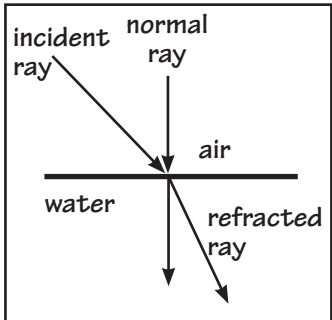
Convex mirrors

Convex mirrors are sometimes called diverging mirrors because when the light strikes the surface, it does not pass through a focal point. The rays will each reflect according to the Law of Reflection. Convex mirrors, like lenses, produce virtual images. Magnifying mirrors use this principle.



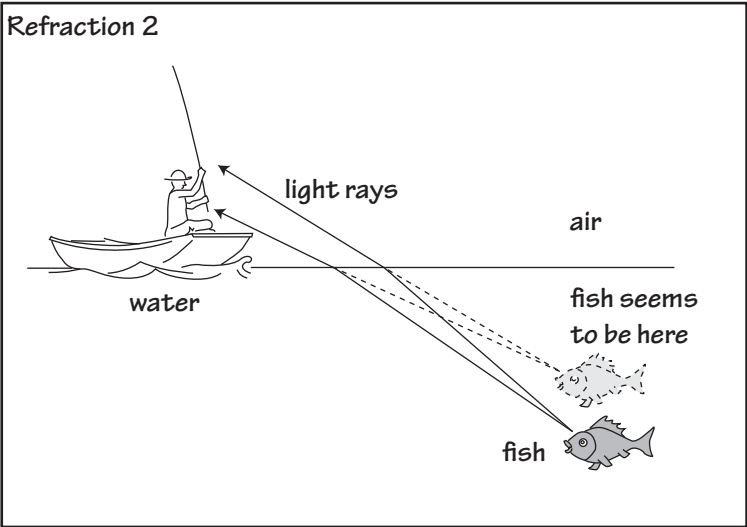
Refraction

6 Study the diagrams at the right, labelled Refraction 1 and Refraction 2. Use this information to write a definition of refraction, and describe how it works.

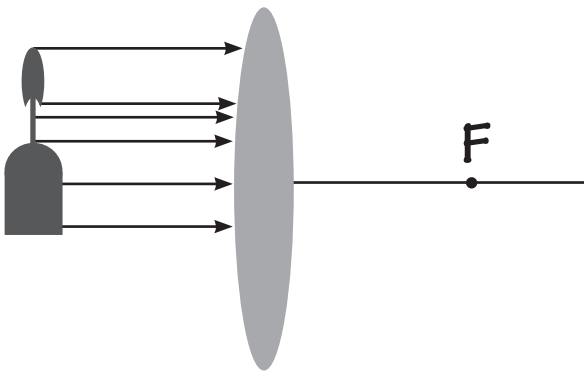


Refraction 1

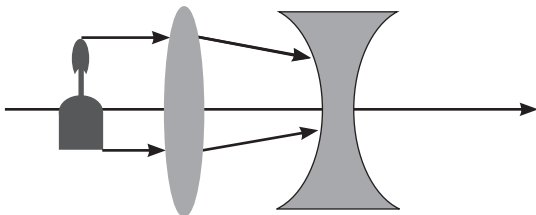
7 Explain the difference between reflection and refraction.



8 Complete the following diagram and state whether the image is seen upside down or right side up.



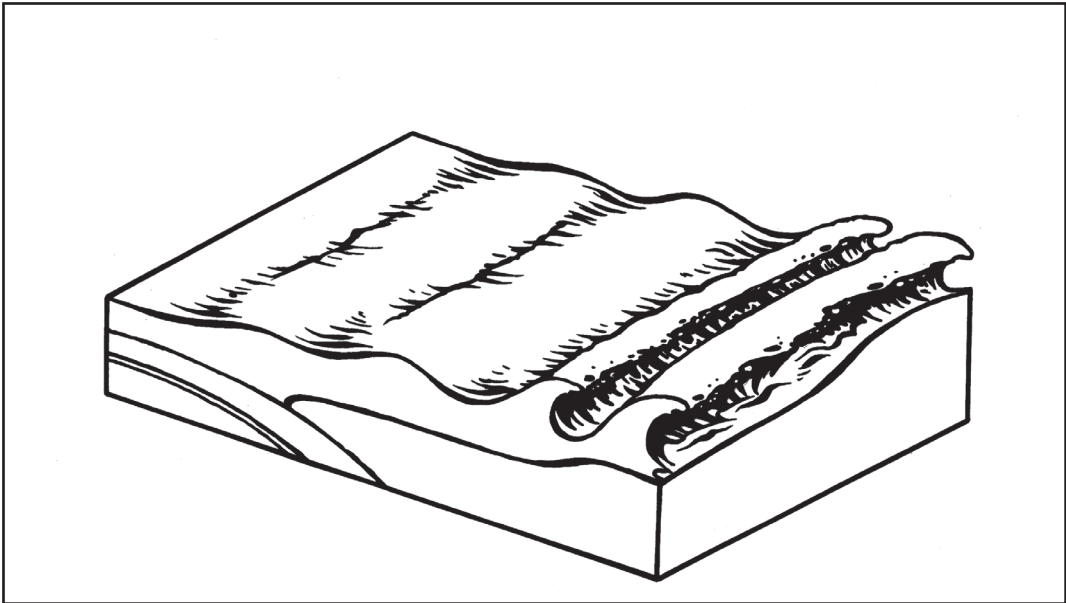
9 Complete the diagram and state whether the image seen will be larger, smaller, inverse or the right way up.



Tsunamis

‘Tsunami’ is a Japanese word meaning *harbour wave*. Now tsunami refers to a large water mass that has a very high energy level, and causes devastation in low-lying areas. The waves are in series, much like the ripples sent out by a pebble dropped into a pond. Earthquakes on the seafloor cause tsunamis. These earthquakes result from plate movement. The sudden slippage causes the water to travel away from the seismic event. Underwater landslides, land slumping into the ocean, volcanic eruptions, nuclear testing underwater and meteorite impacts can also generate tsunamis. The depth of the water controls the speed of a tsunami. As the wave approaches the land, the depth of the water becomes shallower and slows the wave – the rear of the wave is still in deeper water. This causes the water to build up height as well as ‘sucking’ the water from the shoreline.

- 1** Label the diagram with these words:
fault, wavelength, inundation, propagation, seafloor, amplitude, water upthrust, wave generation.



Tsunamis can be generated in several different ways:

- » as a result of land slumping into the ocean, as happened in Lituya Bay, Alaska
- » from volcanic eruptions, such as Krakatoa
- » from nuclear testing, as occurred at Moruroa atoll and Bikini atoll
- » through underwater landslides caused by an earthquake off the coast, as has occurred in Chile
- » through plate movement as occurred off the Sumatran coast in 2004.

- 2 a** Using the information above, write the correct locations in the table.

CAUSE OF TSUNAMI	LOCATION	AMPLITUDE (m)
Volcanic eruption		40.0
Marine nuclear testing		1.5
Plate movement		33.0
Land slumping		3.0
Underwater landslides caused by earthquake		5.0



- b** In the box above right, draw a column graph of amplitude against location. Label and title the graph correctly.
- c** Explain why major plate movement and major eruptions cause high amplitude waves.

3 a Graph the information in the table in the box below.

LOCATION	EARTHQUAKE MAGNITUDE	TSUNAMI AMPLITUDE (m)
Solomon Islands 2007	8.1	5.0
Kuril Islands 2006	8.1	1.5
South Java 2006	7.7	6.0
Sumatra 2004	9.0	33.0
Papua New Guinea 1998	7.1	9.0
South Mariana Islands 1993	7.8	20.0
Alaska 1964	8.3	3.3

b From your graph, what is the relationship between earthquake magnitude and tsunami amplitude?

4 Explain why tsunamis in deep water cause little or no damage, but magnify in intensity as they approach land. Use these words in your explanation: force, amplitude, velocity, mass, depth.

5 Explain how underwater volcanoes can cause a tsunami. (Hint: water is heated very quickly.)

6 In your workbook, draw and annotate a diagram explaining how the coastline and the underwater structures such as reefs, trenches and the continental shelf affect the intensity of a tsunami.

Universe formation - theories

There are several different theories about how the universe began. The Big Bang theory is currently the most accepted theory. It suggests that the Universe was originally concentrated into a single, very small ball. The early universe was so highly unstable that an enormous explosion occurred. This formed the galaxies and stars. Evidence to support this theory is based on the expansion of space, as shown by the redshifts of distant galaxies and cosmic background radiation.

Redshift is the increase in wavelength of electromagnetic radiation. It is an example of the Doppler effect, which shifts the wavelength of emitted light of celestial objects towards the red (longer) wavelengths indicating that the objects are moving away from the Earth due to the continuing expansion of the universe. Redshift is based on the visible electromagnetic spectrum as well as other parts of the spectrum such as gamma rays, x-rays and ultraviolet waves. In 1965, the astronomers Robert Wilson and Arno Penzias discovered cosmic background radiation and showed that this radiation was from the Big Bang.

1 Using the subheadings below, explain the importance of redshift in relation to understanding the Big Bang theory.

Definition _____

Effects _____

Implications _____

The Steady State theory proposes that the universe is the same everywhere and for all time, both in the past and in the future. Scientific calculations suggest that the universe is expanding, so in order to maintain the constant state described in the Steady State theory, matter must be continually being created. Einstein's theory says that mass can be created from energy, as shown in the equation $E = mc^2$. (Energy equals mass multiplied by the speed of light squared)

The Oscillating Universe theory is a variant of the Big Bang theory. It suggests that the universe is expanding and that at some point in the future, it will contract, and that this process of expansion and contraction is continuous. This means that there would be a series of 'big bangs' and 'big crunches', and the universe would be destroyed and re-formed in each cycle.

2 List the essential features of each theory of universe formation.

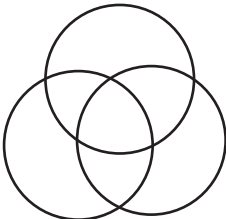
Big Bang _____

Steady State _____

Oscillating Universe _____

3 Draw a Venn diagram to show the points of similarity and difference between the three theories of Universe formation. A model of a Venn diagram for you to copy into your workbook is shown at the right.

4 Why do people develop theories about the origin of the universe? How does this benefit us?



Volcanoes

- 1** Use these words to complete the sentences below: volcano, subduction, strato, summit, island, Sicily, magma, earthquake, Iceland, hotspot, columns, fountains, mantle, Philippines, boundaries, flows, granitic.

There are three basic types of volcanoes: _____, shield and fissure. Each one has its own characteristics. Examples of composite or strato volcanoes are Mt Pinatubo in the _____ and Mt Etna in _____. The layers are composed of ash and lava. These volcanoes occur because of _____ processes and are near plate _____. They are usually very explosive and produce tall eruption _____.

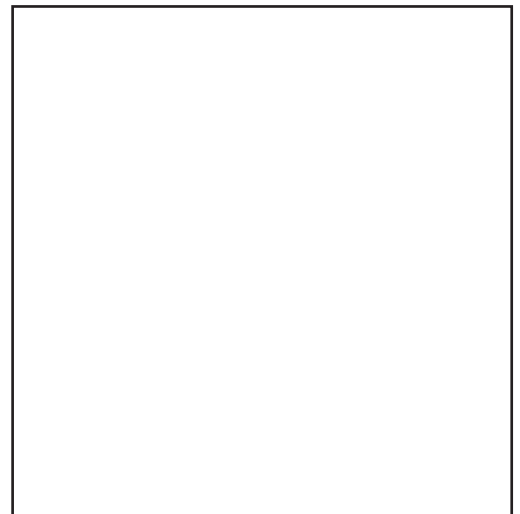
Shield volcanoes occur where there is either a hotspot or a weakness in the crust that allows basaltic magma to rise to the surface. Mauna Loa in Hawaii is the largest _____ on Earth. From its marine base to the _____ is 17 km. Hawaii is 3200 km from a plate boundary, and is the result of a _____. The magma rises through the surrounding rock to form a seamount that finally becomes a volcanic _____.

Fissure volcanoes occur through _____ activity forming a fissure that penetrates below the crust, allowing the magma to rise to the surface. When the magma reacts with ground and surface water, the eruption is explosive. It then changes to a steady turbulent stream of fragmented _____ and magmatic gases that is often toxic, such as in the Laki fissure in _____. As the groundwater decreases, tall eruption columns and tephra mounds are produced. It then changes to effusive basaltic _____, often producing _____ of lava.

Magma that wells up from the _____ but does not reach the surface often produces dykes or vertical walls. Horizontal walls are called sills. Magma also produces large areas of _____ rock called batholiths.

- 2** Identify the three main types of volcanoes.

- 3** In the box at the right, draw and label a diagram of a composite (strato) volcano. Use these labels: vent, crater, sill, dyke, eruption column.



As well as being destructive in the short term, volcanoes can also alter the climate because of the amount of ash and dust that is generated and dispersed into the atmosphere. The lava, however, produces fertile soils on which agriculture depends. Volcanoes can also be a source of income from tourism, as occurs at Mt Agung in Indonesia and the Hawaiian volcanoes. Volcanoes are also responsible for producing the different gases in our atmosphere, except for free oxygen which comes from plants.

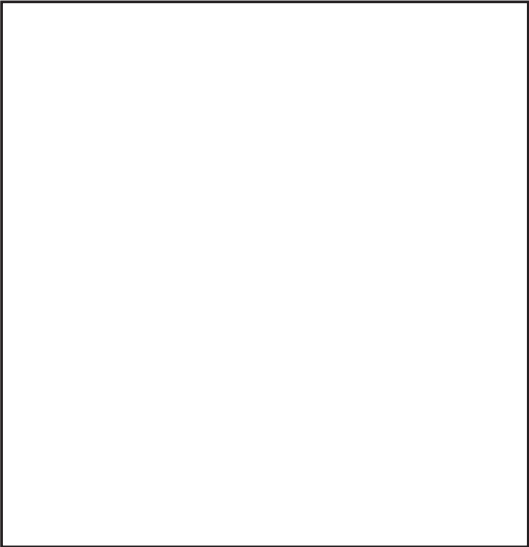
Effusive eruptions, as seen in Iceland and Hawaii, are caused by a small amount of gas and water being in the magma. The water comes from the oceans as the plate is being subducted. As the magma reaches the surface, the pressure is reduced and the gas escapes gently. In an eruption where there is high gas pressure, high temperatures and water within the magma, if the pressure is suddenly released an explosive eruption will occur.

In Iceland, volcanic areas are used not only for tourism but also for the production of geothermal energy to grow food and heat water and houses.

- 4** Explain the eruption process using the analogy of two soft drink bottles. One has been partially undone before opening it fully and the other one has been sealed, shaken and then opened.

5 Using the following data, draw a pie chart of the gases produced by basaltic volcanoes. Group gases which are less than 2%.

+ water (water vapour) 80%	+ hydrogen chloride 0.3%
+ sulfur dioxide 11.5%	+ hydrogen fluoride 0.45%
+ nitrogen 3.4%	+ carbon monoxide 0.2%
+ hydrogen 1%	+ rare and other gases 2.7%
+ carbon dioxide 0.45%	



6 a Why does water vapour have the highest percentage in an eruption?

b Explain how the gases get into the magma.

c Identify the two major greenhouse gases that come from a volcano.

When the Earth first formed, there was a very thin crust and the mantle was very hot and fluid. Many gases escaped through the crust, forming the early atmosphere in a process called outgassing. Eventually, volcanoes formed and emitted most of the gases that are in our atmosphere today. Oxygen did not appear until 3.5 billion years ago when cyanobacteria first started to photosynthesise. Evidence of this comes from stromatolite fossils. Free oxygen is excreted from plants as a by-product.

7 Draw and label a series of diagrams to explain how the atmosphere was formed and how free oxygen appeared in the atmosphere.

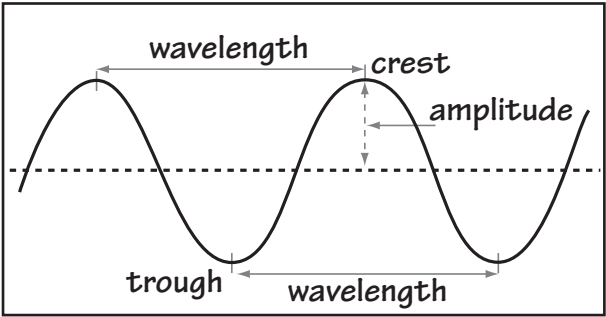
8 Complete the table to show the advantages and disadvantages of living in a volcano-prone area.

	ADVANTAGE	DISADVANTAGE
Agriculture		
Tourism		
Geothermal energy		
Mining		
Health		
Economics		

Waves

All waves have common properties:

- » wavelength (λ) – the distance between two identical points such as crest to crest or trough to trough
- » amplitude – the height from the midpoint to its highest point or its lowest point
- » frequency (f) – the number of waves passing through a fixed point each second, measured in Hertz (Hz)
- » period – the time taken for one wavelength to pass a given point.



Waves are energy carriers.

1 a Measure the amplitude and wavelength of each wave in the table below.

DIAGRAM			
AMPLITUDE			
WAVELENGTH			

b Write the correct description above each diagram in the table above: low frequency, high frequency, and medium frequency.

2 Write a general statement relating wavelength and frequency.

Waves are used in many applications, such as communication, medicine, navigation, radar, sonar, radio and TV. Waves are classified according to their frequency, as you can see in the box below.

+ ELF – extra low frequency	+ MF – medium frequency (AM radio)	+ UHF – ultra high frequency (TV and GPS)
+ VLF – very low frequency (HAM radio)	+ HF – high frequency (CB)	+ microwaves (mobile phones)
+ LF – low frequency (navigation)	+ VHF – very high frequency (FM)	

Information is transmitted on a carrier wave as either AM – amplitude modulation and FM – frequency modulation.

3 Complete the table at the right using the above information.

FREQUENCY NAME	USE
ELF	Submarine communication
	HAM radio
	Navigation
MF	
	CB and FM radio
VHF	
UHF	TV and GPS navigation
Microwaves	

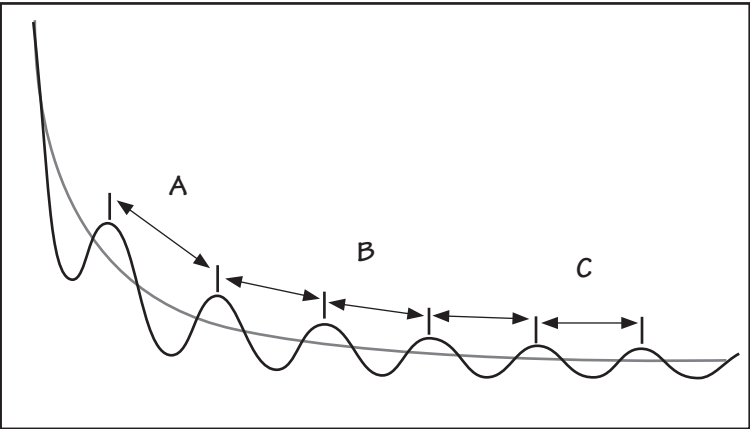
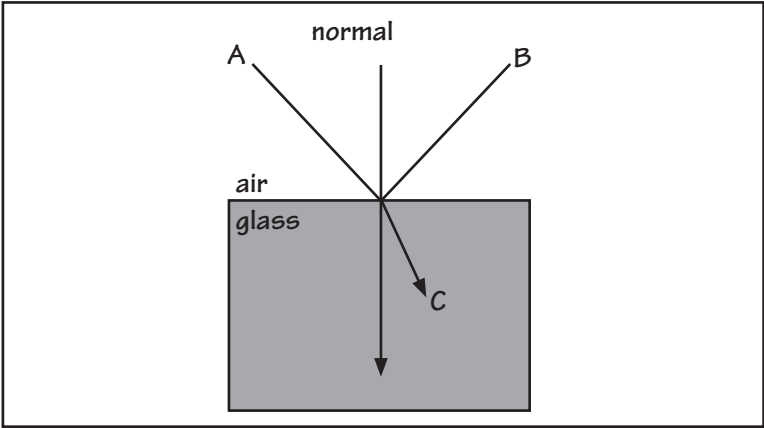
Wave facts

- + The speed of a wave is its frequency multiplied by its wavelength, $v = f\lambda$.
 - + Waves change speed when they pass from one medium to another.
 - + If a wave strikes a surface at an angle, it changes direction. This process is called refraction.
 - + When waves travel through a medium, the amplitude changes. This is called absorption.
 - + Reflection occurs when a wave strikes a material and bounces away from the medium.
- + The wavelength changes as a wave enters a medium.
 - + How a wave reflects is dependent on what angle it strikes the medium.
 - + Wave diffraction occurs as the wave either passes through an opening or moves around an object.
 - + A wave striking the medium at right angles is called a normal ray.

- 4 Draw a mind map in your workbook to organise the information shown in the wave facts box above.

5 Using these terms, write the correct labels at A, B and C on the diagram at the right: incident wave, refracted wave, reflected wave.

6 Measure the wavelengths and amplitudes at points A, B and C in the absorption graph below. Record your measurements in the table at the right.



POINT	A	B	C
WAVELENGTH			
AMPLITUDE			

- 7 Referring to your answers to the previous question, explain briefly what is occurring to the velocity, wavelength and amplitude during absorption.

a Velocity_____

b Wavelength_____

c Amplitude_____

X-rays and atomic particles

X-rays

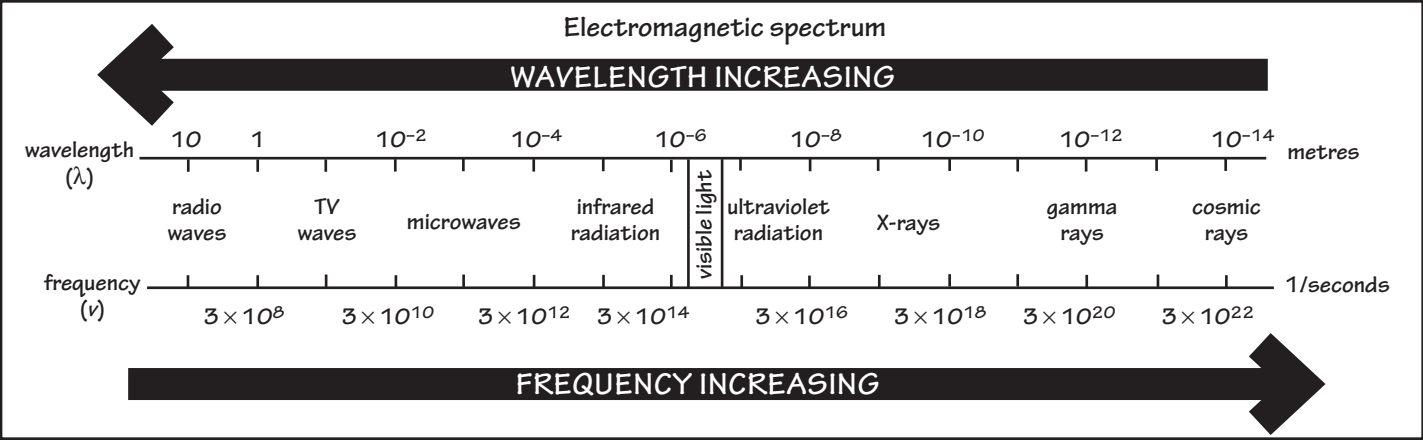
Wilhelm Conrad Roentgen discovered X-rays in 1895. The rays were called X-rays because they were an unknown radiation that came from a vacuum tube. X-rays are produced when a high speed stream of electrons strikes the anode in a cathode ray tube. X-rays are the part of the electromagnetic spectrum that has wavelengths between 0.3 and 0.03 nanometres. They have very high energy. Many objects in space such as black holes, supernova remnants and stars emit X-rays. X-rays are also used to detect broken bones and damaged tissues in the body. X-ray detectors collect photons (a unit of electromagnetic radiation) of X-ray light, which is useful in astronomy. This collection has to be done in space as the Earth's atmosphere stops their penetration.

1 a In your own words, define X-rays and summarise their uses.

b What can happen if X-rays are absorbed by organisms?

Gamma rays

Gamma rays (γ) have the smallest wavelengths, ranging from 0.03 to 0.003 nanometres, and have the most energy of any other wave in the electromagnetic spectrum. Henri Becquerel discovered gamma rays and beta particles. Supernova explosions, nuclear fission, nuclear explosions, black holes and pulsars produce gamma rays. Because of their short wavelengths, gamma rays can kill living cells. Gamma rays are used to sterilise hospital equipment that would be damaged by heat. Both gamma rays and X-rays are part of the electromagnetic spectrum.



- 2 Complete the sequence of electromagnetic radiation.
Radio | Microwaves | Infrared | Visible light | UV | _____ | _____ | _____
- 3 Use these words to complete the following sentence: highly, gamma, stopped.
_____ rays are _____ energetic and can be almost completely _____ by thick sheets of lead and concrete.
- 4 Identify the highest and lowest wavelength rays on the diagram above. What effects could they have on a human?
- 5 Complete the sentences using these words: gamma ray, energetic, energetic.

The smaller the wavelength, the more _____ it becomes. The longer the wavelength, the less _____ it becomes. The most energetic wave in the electromagnetic spectrum is the _____.

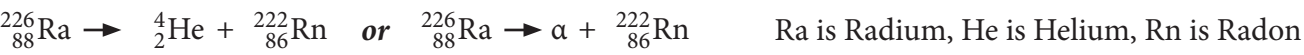
Alpha and beta particles

Alpha and beta particles both have high energy. They are not rays, but arise from unstable and decaying nuclei of atoms. Alpha and beta particles are both forms of radiation.

Alpha particles

Ernest Rutherford identified alpha particles in 1899. An alpha particle (α) is a helium nucleus. Alpha particles will not penetrate human skin, but if swallowed or inhaled will cause mutations of the body’s cells or kill the cells. Alpha particles are heavy and can be stopped by a sheet of paper or a layer of air greater than 3 centimetres.

The chemical reaction of an isotope of radium emitting an alpha particle is:



The top number, e.g. 226, refers to the mass number (protons + neutrons). The lower number, e.g. 88, refers to the atomic number or number of protons.

Alpha particles are used in measuring thickness of paper, foil and smoke detectors. Smoke detectors have a steady electrical current and the detector emits alpha particles, causing ionisation of oxygen and nitrogen. Smoke absorbs alpha particles and this causes a decrease in the electric current which sets off the alarm.

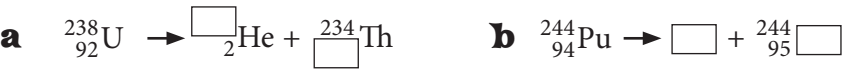
6 Which would you use to sterilise food – X-rays, alpha particles or gamma rays? Explain why.

Beta particles

Beta particles ($_{-1}\beta$) are high speed electrons given off by unstable atoms to restore the energy balance of the atom. Beta particles can be stopped by aluminium foil or by 3 metres of air. Beta particles penetrate the human body and cause chemical changes in cells, which lead to mutations and permanent genetic change. Beta particles are used to treat eye and bone cancer. The chemical reaction of an isotope of uranium emitting a beta particle is:



7 Complete the following equations by writing a number or symbol in each box.



8 Complete the following table about X-rays, gamma rays, alpha and beta particles.

	X-RAYS	GAMMA RAYS	ALPHA PARTICLES	BETA PARTICLES
Penetration (cm)				
Wavelength (nm)			NA	NA
Source				
Uses				

Y chromosomes

If you are a female, you have two X chromosomes. If you are a male, you have an X chromosome and a Y chromosome. Von Nägeli first identified chromosomes in 1842. The word *chromosome* comes from the words *chromo* meaning 'colour' and *soma* meaning 'body'. All organisms have chromosomes in their cells.

Chromosomes are made up of DNA (Deoxyribonucleic acid) molecules, which carry the inherited characteristics called genes. The number of chromosomes varies in each species. Chromosomes are organised into pairs, and range from largest to smallest; this is called a karyotype.

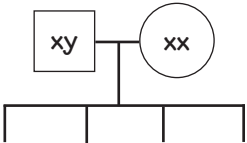
- 1 Chromosomes carry different types and numbers of genes. Hypothesise why butterflies and kangaroos have such different numbers of chromosomes.

ORGANISM	NO. CHROMOSOMES
fruitfly	8
human	46
gorilla	48
goldfish	104
butterfly	380
kangaroo	12
horse	64
elephant	56

Sex chromosomes are the last two chromosomes, and are the smallest chromosomes.

All eggs carry an X chromosome, so the father determines the sex of the offspring, as sperm cells carry either an X or a Y chromosome.

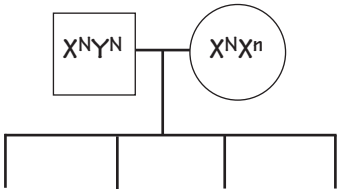
- 2 a Draw a Punnet square to determine the sex of the offspring if a couple has four children.



- b State the sex and percentage of the offspring. _____

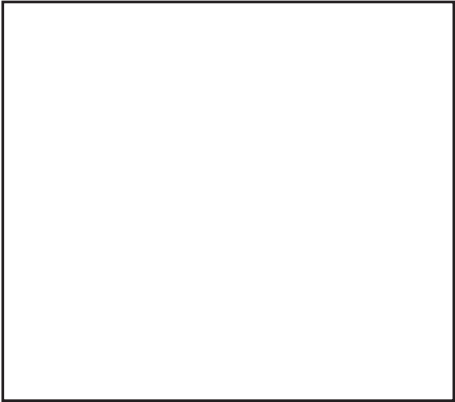
The 45th chromosome in humans is the X chromosome. It often carries inheritable diseases which usually only affect males, such as haemophilia, red-green colour blindness and fragile X.

- 3 Determine the percentage of daughters and sons that are not affected by the inherited disease labelled Xⁿ. Draw a Punnet square to help you find the answer.

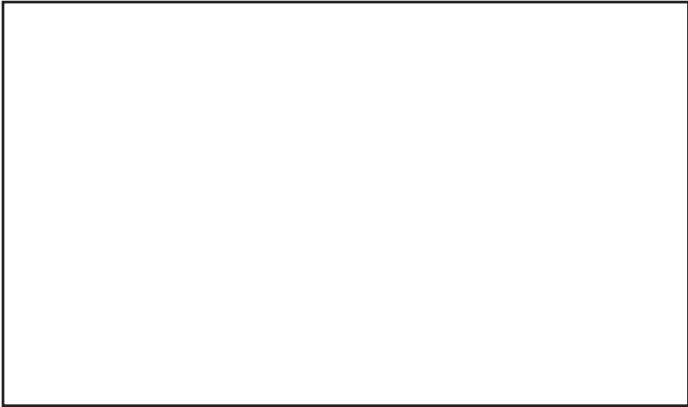


The Y chromosome carries 2% of inheritable genes. Occasionally a male has an XXY condition, known as Klinefelter's syndrome, where an extra X chromosome is inherited. People who suffer from this do not reach sexual maturity and are sterile. A similar condition called Turner's syndrome or Trisomy X exists for females who inherit an extra X chromosome or XXX, which leaves the female sexually under-developed and sterile.

Down syndrome is a chromosomal mutation where there is an extra chromosome, usually chromosome 21 – called trisomy 21, meaning three number 21 chromosomes. Down syndrome can occur in any pregnancy but at the age of 40 years the probability is 1 in 60, and after the age of 40, the probability is significantly increased. Down syndrome occurs in 1 in 600 births. Klinefelters occurs in 1 in 500 births. Super males XYY occur in 1 in 3300 and super females in 1 in 800 births.



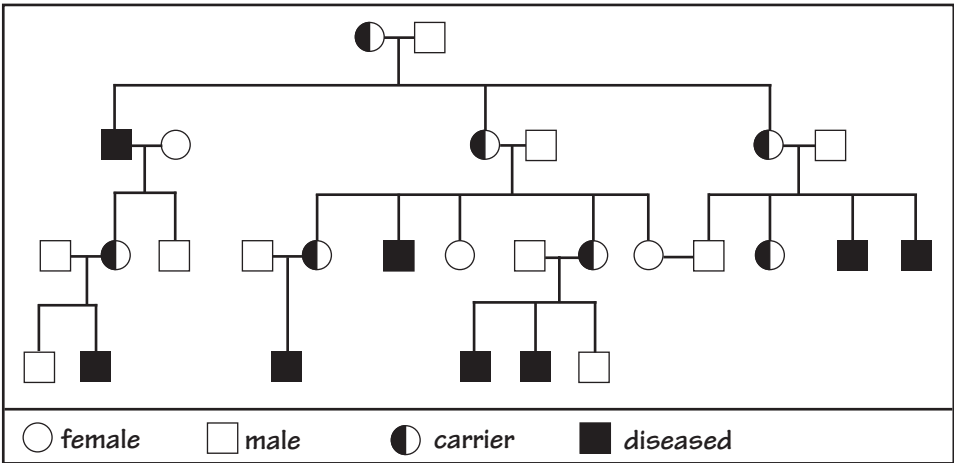
- 4 Using the above information, draw a pie chart showing the incidence of chromosomal diseases in the box at the right.
- 5 Draw a line graph in the box below to show the information in the table at the right.



Chromosomal diseases

AGE OF MOTHER (years)	INCIDENCE PER 1000 BIRTHS
20	1
25	1
30	2
35	3
40	10
45	36
50	82

- 6 From the mutations described above, explain how these changes occur. These terms could be helpful: chemicals, age, X-rays, imperfect copies, spontaneous, meiosis, chromosome change.



- 7 Examine the pedigree family tree above. How is the disease inherited? Give reasons to support your answer.

Zygote and reproduction

A zygote is the result of a sperm cell fertilising an ovum or egg. Reproduction in organisms can occur by budding, fragmentation, fission and spores. Fission occurs in bacteria and protozoa. Fragmentation occurs in planaria worms, budding in seaweeds and spores in ferns and fungi.

1 Complete the table by listing three organisms that reproduce by each type of reproduction.

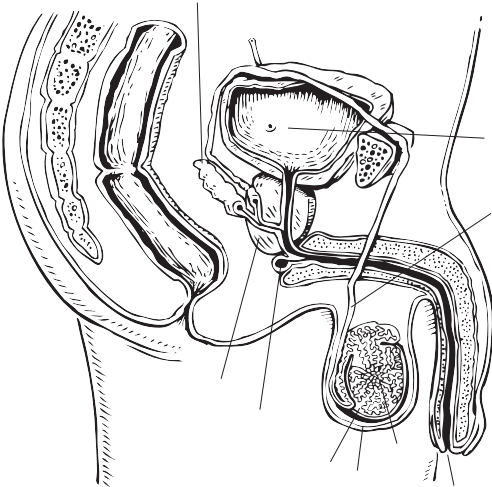
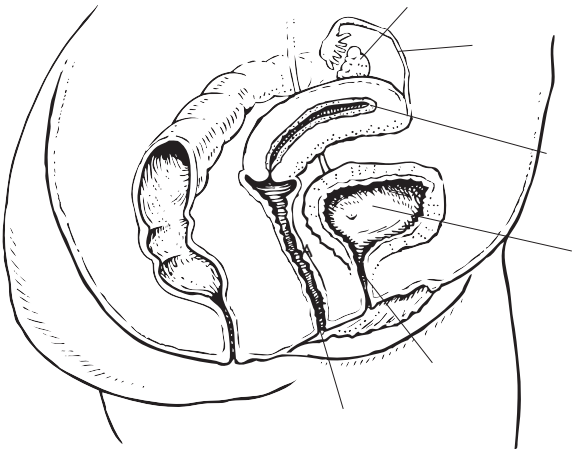
	FISSION	FRAGMENTATION	BUDDING	SPORES	SEXUAL
Organism					

In sexual reproduction in flowering plants and animals, the gametes are formed in the ovaries and testes or anthers.

2 Complete the table to equate plant sexual reproductive components to human reproductive components. Then identify each component as male or female.

3 Use these words to label the diagrams below: vas deferens, uterus, testis, fallopian tube, seminal vesicle, bladder, Cowper's gland, vagina, penis, prostate, urethra, epididymis, ovary, scrotum.

PLANT COMPONENT	HUMAN COMPONENT	MALE OR FEMALE?
Anthers		
Filament		
Stigma		
Style		
Ovary		



- 4 Mark each statement as true or false, then correct the false statements so that they are true.
- a Fertilisation occurs in the vagina. _____
 - b Sperm cells are stored in the epididymis. _____
 - c The prostate and Cowpers glands secrete a fluid to assist sperm survival and mobility. _____
 - d The ovum is viable for 2 – 7 days. _____
 - e Sperm cells are not affected by heat. _____
 - f Ovaries are connected to the fallopian tubes. _____

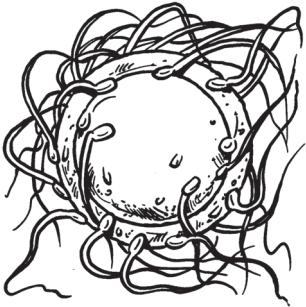
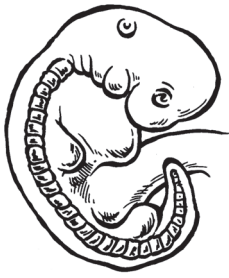
In human reproduction, the ovum is released during ovulation. This occurs between days 12 and 16 of the menstrual cycle. If the ovum is fertilised by a sperm cell, the zygote implants in the uterus around day 21. If implantation does not occur then menstruation occurs on days 1–3 of the cycle.

- 5 Use the information above to label the menstrual cycle timeline.



An ovum is usually fertilised in one of the fallopian tubes. On fertilisation, the ovum has a full complement of chromosomes – half from the mother and half from the father. The zygote begins to grow and make specific cells that will become cells such as nerves, muscles and blood. At the time of implantation, the zygote is called a blastocyst. This becomes the embryo and after 8 weeks the embryo becomes a foetus. Human pregnancy takes 40 weeks. For elephants, it takes up to 18 months!

- 6 Write the name of each stage of development below the correct diagram. Then highlight the section that is developing into the head or spinal cord.

Occasionally on fertilisation, a single ovum and sperm cell divide completely into two. The chromosomes replicate to form two identical zygotes that develop into identical twins. Non-identical twins or fraternal twins develop from the release of two ova fertilised by two different sperm cells, and so the characteristics of these twins will be different.

- 7 Complete the table below using these terms: identical chromosomes, single ovum and sperm, two sperm and two ova, chromosome variation, different physical characteristics, identical physical characteristics.

IDENTICAL TWINS	FRATERNAL TWINS

Selectd answers

Atoms

- 1** Billiard ball, Dalton; Plum pudding, Thomson; Planetary, Rutherford; Shell, Bohr
- 2** **a** F
b T
c T
- 4** **a** F
b T
c F
d T
e T
- 6** carbon, 6, 2; sodium, 11, 3; sulfur, 16, 3; potassium, 19, 4
- 7** 23, 11, 12, 11, sodium
27, 13, 14, 13, aluminium
14, 7, 7, 7, nitrogen
40, 18, 22, 18, argon
9, 4, 5, 4, beryllium

Biotechnology

- 1** 7000 BC – baking bread; 1590 – microscope was invented; 1928 – first antibiotic; 1953 – DNA; 1972 – DNA composition of chimpanzees and gorilla; 1980 – recombinant DNA; 1994 – first GM food; 1997 – cloning a sheep named Dolly; 2000 – Human Genome Project; 2003 – first biotech pet; 2007 – monoclonal antibody work

Chemical reactions

- 1** **a** sulphate
b zinc oxide
c oxygen
- 2** **a** oxygen
b oxide
c dioxide
- 3** **a** sulphate
c hydrogen gas
- 4** **a** potassium nitrate
b Cl
- 5** **a** hydrogen gas
b carbon dioxide
c water

DNA

- 3** GCA GTA ATC ACT CTA
- 4** **a** serine
b isoleucine
c lysine
d tyrosine
e asparagine

f glutamine

g valine

h threonine

- 8** It controls the production of proteins within the cell, which control all chemical processes within the cell.

Electricity

- 1** positive charge, polarised, negative charge, neutral
- 2** induction, contact, friction
- 3** e.g. lightning strike
- 4** static electricity
- 9** 0.5, 1.0, 2; 1, 4, 4; 2, 4, 2

Hormones and nerves

- 1** **a** T
b T
c T
d F
e T
- 3** diabetes
- 4** Cells that respond to a specific hormone
- 6** neurones, axon, dendrites, dendrites, neurotransmitters, myelin

Indicators and pH

- 1** cresol red, thymol blue, phenolphthalein
- 2** **a** basic
b phenol red or cresol red
- 3** **a** azolitmin
b to gain a reading from a wider range of substances
- 4** **a** methyl yellow
b to get a more precise reading within that range

Joules, watts, amps and volts

- 1** The total amount of energy within a closed system remains constant; energy is not created or destroyed
- 2** 15
- 3** **a** 72
b 10
c 12.5
- 4** A joule is one watt/second

Motion

- 1** **a** 80
b 72 m/s
c 450 m
- 3** **a** BC, EF
b FG
c DE
d AB
- 4** **a** 11 m/s
b 4 m/s
c 12.3 m/s
d 11 m/s
- 6** **a** 4.2 m/s²
b 630 N
- 7** 1.6 m/s
- 8** 182 m
- 9** 14 m/s
- 10** 2.25 N
- 11** **a** -8 m/s²
b 3.75 s
c 168.75 m

Natural events

- 3** Low pressure develops off shore, strong winds, torrential rain, erosion, habitat loss, loss of agricultural produce, destruction of houses, human emotional distress, appointment of General Cosgrove to administer the reconstruction.

Newton's Laws

- 1** **a** The loose objects remain moving as they haven't yet been acted on by the forces that are slowing the car
- 2** **a** people may fall backwards
b people may fall in the direction opposite to which the bus is turning
c people may fall backwards
- 3** **a** 10, 5, 20; 80, 4, 20; 90, 6, 15; 20, 4, 5; 36, 6, 6
b 4 m/s²
c 1500 N
d 85 kg

Non-renewable resources

- 1** the Sun
- 3** Sun, plants, buried, compression, coal
- 4** **a** e.g. making plastics, nylon and polyester; lubrication for moving parts of machinery.
b e.g. electricity generation, steel production, cement

manufacture.

- 5** Produces carbon dioxide, which contributes to the increasing temperature of the planet.
- 6** **a** energy
b electricity
c fossil fuel
d oil
e coal
f combustion

Nuclear fission and fusion

- 1** **a** A neutron is fired at the nucleus of a uranium-235 atom, which splits into a xenon atom, a strontium atom, 2 neutrons and energy.
b superscript – mass number; subscript – atomic number
- 2** A neutron is fired at the nucleus of a uranium-235 atom, which splits into a barium atom, a krypton atom, 3 neutrons and energy.
- 3** deuterium + tritium → helium + neutron

Periodic Table

- 1** (In column order)
2.8.8, 8, 3, argon
2.1, 1, 2, lithium
2.4, 4, 2, carbon
2.8.7, 7, 3, chlorine
2.8.8.1, 1, 4, potassium
- 2** **a** 2.1
b 2.8.8.2
c 2.8.2
d 2.3

Quasars and stars

- 1** **a** To detect the radio waves emitted by them
b It is continuing to expand
- 3** **a** neutron stars are formed in the collapse of massive stars; white dwarfs are formed in the collapse of low-mass stars.
c black hole
- 4** deuterium + tritium → helium + neutron + energy

Reflection and refraction

- 3** **a** converging
- 4** **a** The real image is smaller and upside down
b The eye sees things upside down, and the brain interprets the images 'right way up'.
- 6** Light can be 'bent' as it passes through different mediums (refracted). Refraction is the changing of the angle of a light wave as it passes from one medium to another medium

e.g. air to water. Refraction occurs at the boundary of the medium with a change in speed. The speed of a wave is dependent upon the optical density through which it moves. A ray of light going through the air will refract toward the normal if it strikes glass. If the light ray moves from a slow to fast medium it will bend away from the normal.

- 7** Reflection – light bounces off a surface. Refraction – light passes into the new medium

Universe formation

- 2** Steady State – matter is continuously created, density remains constant
Oscillating Universe – big crunches and bangs, cycle is never ending
Big Bang – continuously expanding, galaxies are moving away

Volcanoes

- 1** strato, Philippines, Sicily, subduction, boundaries, columns, volcano, summit, hotspot, island, earthquake, magma, Iceland, flows, fountains, mantle, granitic
2 strato, shield, fissure

Waves

- 1** **b** low frequency, medium frequency, high frequency
2 The greater the wavelength, the lower the frequency
5 A – incident wave; B – reflected wave; C – refracted wave
7 **a** decreasing
b increasing
c decreasing

X-rays and other rays

- 2** X-rays, gamma rays, cosmic rays
3 Gamma, highly, stopped
4 highest – radio waves; lowest – cosmic rays
5 Energetic, energetic, gamma
6 gamma rays
7 **a** 4, 90
b $_{-1}^0\beta$, Am

Y chromosome

- 2** **a** 50% female, 50% male
3 25% not affected female; 25% affected female; 25% not affected male; 25% affected male
7 carrier mother to son

Zygotes and reproduction

- 4** **a** F
b T
c T
d T
e F
f F
6 zygote, blastocyst, embryo, foetus
7 identical twins: identical chromosomes, single ovum and sperm, identical physical characteristics
fraternal twins: two ova, two sperm; chromosomal variation; different physical characteristics

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing **the highest quality learning outcomes with instant usability.**

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

A–Z of Science Book 4

This series of photocopiable books extends across all branches of this subject area, including biology, chemistry, astronomy, earth and environmental science. The activities include basic revision of fundamental concepts as well as ideas and tasks that will challenge students to think beyond recalling facts. Each topic is designed to engage student interest in science, as well as consolidate and extend their scientific knowledge and understanding. Book 3 is suitable for years 7 and 8, while Book 4 is more advanced and will suit years 9 and 10.

Topics include	
Atoms Biotechnology Chemical reactions Earthquakes Fossils Genes Hormones and nerves Indicators and pH	Motion Non-renewable resources Periodic Table Quasars and Stars Tsunamis Volcanoes Waves X-rays and atomic particles

Associated titles in the Instant Lessons series

- A to Z of Science (Book 3)
- Legends of Science (Books 1, 2, 3, 4)
- Problem Solving in Science (Books 1, 2)
- Scientific Enquiry Skills (Books 1, 2)
- Experimenting with Science
(The living world, The physical world, Forces and electricity)
- Achieve! Science
 - Chemical reactions, materials and particles
 - Energy, electricity and movement
 - Life processes and the environment
 - Scientific investigation, plants, rocks and outer space

Visit the Blake Education website or email info@blake.com.au for more information about the Instant Lessons series, other science titles, the full list of contents for all titles and ordering advice



Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

